claude-wsl-ubuntu / 4f364eb7-a600-4a2c-9b37-e5a313b6db9d

Metadata

- Source: `claude-wsl-ubuntu`
- Kind: `claude`
- Source file: `\\wsl.localhost\Ubuntu\home\avidullu\.claude\projects\-home-avidullu-projects-khelsutra-guru\4f364eb7-a600-4a2c-9b37-e5a313b6db9d\subagents\agent-a5598ac50ddee6368.jsonl`
- SHA-256: `dce5b4bbcb6ec1e982a63e999cd9103aeeae8833935bbe196540cf0b7cb49a70`
- Source modified: `2026-06-29T12:01:33+00:00`
- Imported at: `2026-07-05T16:48:30+00:00`
- project: `subagents`
- session_id: `4f364eb7-a600-4a2c-9b37-e5a313b6db9d`

Transcript

1. user (2026-06-29T11:52:19.032Z)

You are an isolated worker doing a DEEP, NON-DESTRUCTIVE index + dedup REPORT of a ~1.2 TB GoPro video archive on a freshly-mounted Windows drive (F:) in WSL. This runs independently of other work ? do not touch any git repo, do not delete or move ANYTHING.

HARD CONSTRAINTS (read twice)
- **READ-ONLY on /mnt/f.** NEVER delete, move, rename, or write anything under /mnt/f. No `rm`, `mv`, `mkdir` on /mnt/f. Report + plan only.
- Write ALL your outputs to a dedicated dir OUTSIDE the drive and outside any repo: `/home/avidullu/projects/khelsutra-guru/f_drive_index/` (create it). Nothing else gets written anywhere else.
- Filenames contain spaces, brackets `[Khelsutra]`, and mixed case. Always null-delimit (`find -print0`, `read -d ''`) and quote paths. Do not let one weird filename abort the run.
- Checkpoint progress to disk frequently so a long run is resumable and partial results survive.

WHAT / WHERE
The archive is on /mnt/f (mountpoint confirmed). The primary data folders:
- `/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos` (the main one)
- `/mnt/f/[Khelsutra] GoPro Hero Black 12/` (parent ? check siblings)
- `/mnt/f/Data/Initial GoPRo with Sports Recordings`
FIRST do a quick top-level survey of /mnt/f (one or two levels, sizes via `du -sh` on candidate dirs) to confirm where the video data actually lives, then deep-index ALL data-bearing folders you find (ignore `$RECYCLE.BIN`, `System Volume Information`, `.Spotlight-V100`, `.fseventsd`).

GOAL
1. **Catalog** every media file: full path, size (bytes), mtime, extension, category. Categories: VIDEO (.mp4/.MP4/.mov/.MOV/.avi/.mkv/.m4v/.insv/.360), PROXY (.lrv/.LRV ? GoPro low-res), THUMB (.thm/.THM), OTHER. Use `find ... -printf '%s\t%T@\t%p\0'` style enumeration. Save the raw catalog to `f_drive_index/catalog_raw.tsv`.
2. **De-duplicate by content** (the repo's canonical id is `video_id = full-file MD5`, per backend/utils/hashing.py::compute_video_id). Be efficient on 1.2 TB ? staged:
   a. Group VIDEO files by exact byte size. Distinct size ? distinct content ? unique (no full hash needed yet).
   b. For each same-size group (?2 files), compute a cheap partial signature: `md5( first 16MB ++ last 16MB )` via `dd`. Split groups by (size, partial-sig).
   c. For any (size, partial-sig) collision group still ?2, compute the FULL `md5sum` to CONFIRM exact duplicates. Confirmed dups share the full MD5 = their video_id.
   d. Also compute full MD5 for any file whose basename matches a known corpus/golden name (list below) so the index cross-references cleanly.
   e. After dedup is done, if time/budget remains, progressively full-MD5 the remaining unique VIDEO files too (checkpointing), so the index carries real video_ids; for any not yet fully hashed, record a provisional id `"size:<bytes>:p:<partialsig>"` and mark `"video_id_provisional": true`.
3. **Metadata** (cheap, header-only): if `ffprobe` is available, capture duration_s, fps, width, height, codec per VIDEO file (fast ? don't decode frames). Skip gracefully if ffprobe absent.

4. **Cross-reference** against the cloud corpus `gs://khelsutra-rally-corpus` (gcloud is authed as the owner; `gcloud storage ls -l gs://khelsutra-rally-corpus/videos/` works). For each F: video, note: does a same-name and/or same-size object already exist in the bucket? The 15 golden clip names are: adarsh_avi_singles, kushagra_singles, largetest_doubles, gbaaddy, testlarge_short, mahadevpura_singles, mahadevpura_2, GX010128, mahadevpura_1, GX030094, Badminton_BXH_2, GX010141, GX010137, Boxhill_Doubles, GX020094 (GoPro originals are named like GX0101NN.MP4). Produce a **migration-gap list**: which golden/corpus videos exist on F: but are NOT yet in `gs://khelsutra-rally-corpus/videos/` (this closes the known DATA_IN_GCS §5 gap where 14/15 corpus videos aren't migrated). NOTE: `gcloud storage cp` treats `[...]` as a glob, so any future upload of these bracketed paths must stage through a bracket-free temp path ? mention this in the plan, but DO NOT upload anything yourself.

OUTPUTS (write to /home/avidullu/projects/khelsutra-guru/f_drive_index/)

- `catalog_raw.tsv` ? every media file (path, size, mtime, ext, category).
- `master_video_index.json` ? array of unique VIDEO entries: {video_id (or provisional), video_id_provisional, path, size_bytes, mtime, ext, duration_s, fps, width, height, codec, dup_group_id, is_canonical, in_gcs (bool/uncertain), gcs_object (if matched), golden_name (if matched)}. This is the artifact "any repo can refer to."
- `dedup_report.md` ? human-readable: total files & bytes scanned; # unique videos; duplicate GROUPS (each: the full MD5, the proposed canonical-to-keep [pick earliest mtime; tie-break shortest path], the redundant copies with full paths, bytes reclaimable); total reclaimable bytes; counts of proxies/thumbs. A clear PROPOSED dedup action list ? but execute NONE of it.
- `gcs_migration_gap.md` ? which golden/corpus videos are present on F:, which are already in GCS, which still need upload (with the bracket-glob caveat + a suggested bracket-free staging recipe).
- `progress.log` ? append-only progress + any files that errored (permission/IO), so nothing is silently dropped.

FINAL MESSAGE back to me

A TIGHT summary (this is data returned to an orchestrator, not prose for an end user): total media files & TB scanned; # unique videos vs # duplicate copies; total bytes reclaimable by dedup; how many of the 15 golden clips are present on F: and how many still need GCS upload; whether full-MD5 is complete or partial; and the absolute paths of the 4 output artifacts. Note any folders/files you could not read.

Be methodical, checkpoint often, and remember: READ-ONLY on /mnt/f, report-only, no deletions.

2. assistant (2026-06-29T11:52:21.214Z)

I'll start this deep, non-destructive index of the GoPro archive. Let me begin by setting up the output directory and surveying the drive.

3. user (2026-06-29T11:52:25.628Z)

```
Output dir ready
---MOUNT CHECK---
/mnt/f is a mountpoint
total 17412
drwxrwxrwx 1 root root     512 Aug 22  2021 $RECYCLE.BIN
drwxrwxrwx 1 root root     512 Dec 31  1979 .
drwxr-xr-x 6 root root    4096 Jun 12 06:22 ..
drwxrwxrwx 1 root root     512 Oct 31  2024 .Spotlight-V100
drwxrwxrwx 1 root root     512 Oct 31  2024 .fseventsd
drwxrwxrwx 1 root root     512 Oct 31  2024 Data
-rwxrwxrwx 1 root root 7966967 Jun 27  2020 Install Security for Mac.zip
-rwxrwxrwx 1 root root 8933456 Jun 27  2020 Install Security for Windows.exe
drwxrwxrwx 1 root root     512 Aug 22  2021 System Volume Information
drwxrwxrwx 1 root root     512 Mar  6 08:29 [Khelsutra] GoPro Hero Black 12
```

4. assistant (2026-06-29T11:52:31.537Z)

Mount confirmed. Let me set up my task list and survey the data folders.

5. user (2026-06-29T11:52:36.288Z)

```
Todos have been modified successfully. Ensure that you continue to use the todo list to track
your progress. Please proceed with the current tasks if applicable
```

6. user (2026-06-29T11:52:51.772Z)

```
--- top-level dirs under /mnt/f ---
total 0
drwxrwxrwx 1 root root 512 Mar  6 08:29 .
drwxrwxrwx 1 root root 512 Dec 31  1979 ..
drwxrwxrwx 1 root root 512 Jun  7 06:16 GoPro Recorded Videos
drwxrwxrwx 1 root root 512 Jun 15 18:51 KhelSutra_work
--- Data/ ---
total 1024
drwxrwxrwx 1 root root   512 Oct 31  2024 .
drwxrwxrwx 1 root root   512 Dec 31  1979 ..
drwxrwxrwx 1 root root   512 May 12  2025 Backup
drwxrwxrwx 1 root root   512 May 13  2025 Emails
drwxrwxrwx 1 root root   512 Feb  4 13:18 Initial GoPRo with Sports Recordings
drwxrwxrwx 1 root root   512 May 12  2025 Motorcyling
-rwxrwxrwx 1 root root 58030 Sep 21  2024 Search.pdf
drwxrwxrwx 1 root root   512 May 13  2025 Wife Problem
--- tool availability ---
/usr/bin/ffprobe
/usr/bin/md5sum
/usr/bin/dd
/mnt/c/Users/avidu/AppData/Local/Google/Cloud SDK/google-cloud-sdk/bin/gcloud
ffprobe: ffprobe version 6.1.1-3ubuntu5 Copyright (c) 2007-2023 the FFmpeg developers
gcloud: Google Cloud SDK 571.0.0
```

7. assistant (2026-06-29T11:52:54.531Z)

```
All tools available (ffprobe, md5sum, dd, gcloud). There are more folders than expected:
`KhelSutra_work` sibling, and under `Data/` there are `Backup`, `Motorcyling`, etc. Let me size
up all candidate dirs.
```

8. user (2026-06-29T11:53:03.429Z)

```
--- du -sh on candidate data dirs (this may take a bit) ---
1.1T    /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos
13G     /mnt/f/[Khelsutra] GoPro Hero Black 12/KhelSutra_work
18G     /mnt/f/Data/Initial GoPRo with Sports Recordings
33G     /mnt/f/Data/Backup
16G     /mnt/f/Data/Motorcyling
42M     /mnt/f/Data/Emails
11G     /mnt/f/Data/Wife Problem
--- one level deeper into the two GoPro mains ---
```

GoPro Recorded Videos ###

```
total 3072
drwxrwxrwx 1 root root     512 Jun  7 06:16 .
drwxrwxrwx 1 root root     512 Mar  6 08:29 ..
drwxrwxrwx 1 root root     512 Mar  6 08:30 2026 March 6
drwxrwxrwx 1 root root     512 Jun 17 02:44 Badminton - Boxhill - 17 June
drwxrwxrwx 1 root root     512 Jun  2 02:37 Badminton 1 June 2026
drwxrwxrwx 1 root root     512 Jun 14 02:50 Badminton 12 June
drwxrwxrwx 1 root root     512 Jun 12 05:47 Badminton 12 June 2026
drwxrwxrwx 1 root root     512 May 15 00:59 Badminton 14 May 2026
drwxrwxrwx 1 root root     512 May  4 05:40 Badminton 2 May 2026
drwxrwxrwx 1 root root     512 May 22 05:19 Badminton 22 May 2026
drwxrwxrwx 1 root root     512 Jun 24 23:05 Badminton 25 June 2026
drwxrwxrwx 1 root root     512 May  5 00:39 Badminton 4 May 2026
drwxrwxrwx 1 root root     512 Jun  9 15:03 Badminton 9 June 2026
drwxrwxrwx 1 root root     512 Jun 19 02:45 Badminton Boxhill 19 June
```

```
drwxrwxrwx 1 root root     512 Jun 15 05:06 Bdminton 15 June
drwxrwxrwx 1 root root     512 May 20 03:09 GoPro 20 May 2026
drwxrwxrwx 1 root root     512 May 20 03:31 Highlights Generator Data
drwxrwxrwx 1 root root     512 Jun  3 06:15 June 3 2026
drwxrwxrwx 1 root root     512 May 22 00:35 May 19 2026
drwxrwxrwx 1 root root     512 Jun 15 08:23 Misc
drwxrwxrwx 1 root root     512 Jun 15 07:56 Roora June 15
drwxrwxrwx 1 root root     512 Jun 23 07:00 SHANAN
drwxrwxrwx 1 root root     512 Jun 10 12:53 Tabletennis 10 June
drwxrwxrwx 1 root root     512 Jun 12 05:47 Tabletennis 11 June
drwxrwxrwx 1 root root     512 Jun 13 14:10 Training
-rwxrwxrwx 1 root root 1682294 Jun 15 12:16 roora_batch_log.txt
-rwxrwxrwx 1 root root    5649 Jun 15 12:16 roora_june15_proxy_manifest.csv
```

KhelSutra_work ###

```
total 0
drwxrwxrwx 1 root root 512 Jun 15 18:51 .
drwxrwxrwx 1 root root 512 Mar  6 08:29 ..
drwxrwxrwx 1 root root 512 Jun 19 08:01 19june_proxies
drwxrwxrwx 1 root root 512 Jun 15 18:51 kushagra_june12_proxies
```

Initial GoPRo with Sports Recordings ###

```
total 18285568
drwxrwxrwx 1 root root        512 Feb  4 13:18 .
drwxrwxrwx 1 root root        512 Oct 31  2024 ..
-rwxrwxrwx 1 root root       4096 Dec  1  2023 ._GX010002.MP4
-rwxrwxrwx 1 root root       4096 Dec  1  2023 ._GX010008.MP4
-rwxrwxrwx 1 root root       4096 Dec  1  2023 ._GX010009.MP4
-rwxrwxrwx 1 root root     430646 Nov 29  2023 GL010013.LRV
-rwxrwxrwx 1 root root    1092638 Jun 26  2025 GL010015.LRV
-rwxrwxrwx 1 root root   14874881 Jul 13  2025 GL010016.LRV
-rwxrwxrwx 1 root root    1018760 Dec 31  2015 GL010021.LRV
-rwxrwxrwx 1 root root     100870 Dec 31  2015 GL010022.LRV
-rwxrwxrwx 1 root root     168452 Nov 25  2025 GL010024.LRV
-rwxrwxrwx 1 root root     130730 Nov 25  2025 GL010025.LRV
-rwxrwxrwx 1 root root    3105459 Nov 25  2025 GL010026.LRV
-rwxrwxrwx 1 root root     262973 Nov 25  2025 GL010027.LRV
-rwxrwxrwx 1 root root  261053372 Dec  2  2025 GL010029.LRV
-rwxrwxrwx 1 root root  222153422 Dec  2  2025 GL010030.LRV
-rwxrwxrwx 1 root root    2050861 Dec  2  2025 GL010031.LRV
-rwxrwxrwx 1 root root     144304 Dec  3  2025 GL010032.LRV
-rwxrwxrwx 1 root root   52095434 Dec 30 13:50 GL010038.LRV
-rwxrwxrwx 1 root root    1470784 Dec 30 13:50 GL010039.LRV
-rwxrwxrwx 1 root root  101573660 Dec 30 13:59 GL010040.LRV
-rwxrwxrwx 1 root root    2584190 Dec 30 13:59 GL010041.LRV
-rwxrwxrwx 1 root root     478427 Dec 30 14:00 GL010042.LRV
-rwxrwxrwx 1 root root  129879539 Dec 30 14:11 GL010043.LRV
-rwxrwxrwx 1 root root  174085330 Dec 30 14:29 GL010044.LRV
-rwxrwxrwx 1 root root   62370761 Dec 30 14:41 GL010045.LRV
-rwxrwxrwx 1 root root  486781899 Dec 30 15:15 GL010046.LRV
-rwxrwxrwx 1 root root     452906 Jan  1 08:14 GL010047.LRV
-rwxrwxrwx 1 root root  270464369 Jan 15 04:40 GL010048.LRV
-rwxrwxrwx 1 root root    7239825 Jan 15 04:44 GL010049.LRV
-rwxrwxrwx 1 root root  447624115 Jan 15 05:19 GL010050.LRV
-rwxrwxrwx 1 root root  260620872 Jan 16 02:06 GL010051.LRV
-rwxrwxrwx 1 root root     280295 Jan 16 02:20 GL010052.LRV
-rwxrwxrwx 1 root root   72624997 Jan 16 02:30 GL010053.LRV
-rwxrwxrwx 1 root root     492336 Jan 16 02:30 GL010054.LRV
-rwxrwxrwx 1 root root  646050493 Jan 16 03:38 GL010055.LRV
-rwxrwxrwx 1 root root  433210765 Jan 17 05:27 GL010056.LRV
-rwxrwxrwx 1 root root  322105986 Jan 17 05:50 GL010057.LRV
-rwxrwxrwx 1 root root    1428135 Jan 17 12:05 GL010058.LRV
```

9. assistant (2026-06-29T11:53:07.077Z)

The data is confirmed. Main archive is ~1.1T in "GoPro Recorded Videos". Also note `._GX*.MP4`

```
AppleDouble files (4KB metadata stubs ? not real videos). Now let me build the full catalog.
I'll do the enumeration of all media files across all data-bearing dirs, null-delimited.
```

10. user (2026-06-29T11:53:16.812Z)

```
Total files enumerated: 1795
```

11. assistant (2026-06-29T11:53:19.198Z)

```
1795 files enumerated. Now let me classify them into categories and build the `catalog_raw.tsv`.
I'll write a Python script for robust null-delimited handling.
```

12. assistant (2026-06-29T11:53:32.896Z)

```
#!/usr/bin/env python3
"""Read the null-delimited enumeration, classify each file, write catalog_raw.tsv.
Categories:
  VIDEO  .mp4 .mov .avi .mkv .m4v .insv .360
  PROXY  .lrv  (GoPro low-res)
  THUMB  .thm
  OTHER  everything else
AppleDouble sidecars (basename starts with '._') are flagged in category as OTHER
and excluded from VIDEO dedup even if extension is .mp4 (they are 4KB metadata stubs).
"""
import os, sys, datetime

VIDEO_EXT = {'.mp4', '.mov', '.avi', '.mkv', '.m4v', '.insv', '.360'}
PROXY_EXT = {'.lrv'}
THUMB_EXT = {'.thm'}

infile = '/home/avidullu/projects/khelsutra-guru/f_drive_index/catalog_all_files.nul'
outfile = '/home/avidullu/projects/khelsutra-guru/f_drive_index/catalog_raw.tsv'

with open(infile, 'rb') as f:
    data = f.read()

records = []
for chunk in data.split(b'\0'):
    if not chunk:
        continue
    # split on first two tabs only (path may contain tabs?unlikely but safe)
    try:
        text = chunk.decode('utf-8', errors='surrogateescape')
    except Exception:
        text = chunk.decode('latin-1')
    parts = text.split('\t', 2)
    if len(parts) != 3:
        sys.stderr.write('BAD RECORD: %r\n' % text)
        continue
    size_s, mtime_s, path = parts
    try:
        size = int(size_s)
    except ValueError:
        sys.stderr.write('BAD SIZE: %r\n' % text)
        continue
    mtime = float(mtime_s)
    base = os.path.basename(path)
    ext = os.path.splitext(base)[1].lower()
    is_appledouble = base.startswith('._')
    if is_appledouble:
        cat = 'OTHER'  # AppleDouble metadata stub
    elif ext in VIDEO_EXT:
        cat = 'VIDEO'
    elif ext in PROXY_EXT:
```

```
        cat = 'PROXY'
    elif ext in THUMB_EXT:
        cat = 'THUMB'
    else:
        cat = 'OTHER'
    records.append((path, size, mtime, ext, cat, is_appledouble))
```

Sort by path for stable output

```
records.sort(key=lambda r: r[0])

with open(outfile, 'w', encoding='utf-8', errors='surrogateescape') as out:
    out.write('path\tsize_bytes\tmtime_epoch\tmtime_iso\text\tcategory\n')
    for path, size, mtime, ext, cat, isad in records:
        iso = datetime.datetime.fromtimestamp(mtime).isoformat()
        out.write('%s\t%d\t%.6f\t%s\t%s\t%s\n' % (path, size, mtime, iso, ext, cat))
```

summary

```
from collections import Counter, defaultdict
cat_count = Counter()
cat_bytes = defaultdict(int)
ext_count = Counter()
for path, size, mtime, ext, cat, isad in records:
    cat_count[cat] += 1
    cat_bytes[cat] += size
    ext_count[ext] += 1

total_bytes = sum(r[1] for r in records)
print('Total files: %d' % len(records))
print('Total bytes: %d (%.3f TB)' % (total_bytes, total_bytes/1e12))
print('--- by category ---')
for c in ('VIDEO', 'PROXY', 'THUMB', 'OTHER'):
    print('  %-6s %6d files  %15d bytes  (%.3f GB)' % (c, cat_count[c], cat_bytes[c],
cat_bytes[c]/1e9))
print('--- top extensions ---')
for e, n in ext_count.most_common(20):
    print('  %-8s %5d' % (e or '(none)', n))
```

count AppleDouble

```
ad = sum(1 for r in records if r[5])
print('AppleDouble ._ sidecars (excluded from VIDEO): %d' % ad)
```

13. user (2026-06-29T11:53:33.975Z)

```
File created successfully at: /home/avidullu/projects/khelsutra-guru/f_drive_index/classify.py
(file state is current in your context ? no need to Read it back)
```

14. user (2026-06-29T11:53:39.757Z)

```
Total files: 1795
Total bytes: 1231768920373 (1.232 TB)
--- by category ---
  VIDEO     220 files    1155103142663 bytes  (1155.103 GB)
  PROXY     163 files      33246122309 bytes  (33.246 GB)
  THUMB     163 files          1349996 bytes  (0.001 GB)
  OTHER    1249 files      43418305405 bytes  (43.418 GB)
--- top extensions ---
  .jpg       587
  .pdf       361
  .mp4       222
  .png       184
  .lrv       163
  .thm       163
  .csv        24
```

```
  .zip        21
  .log        13
  .xlsx        7
  .m4a         7
  .mp3         7
  .heic        5
  .xml         5
  .md          5
  .nef         4
  .xls         3
  .nfs13save     3
  .txt         3
  .json        2
AppleDouble ._ sidecars (excluded from VIDEO): 3
--- stderr (bad records) ---
1796 catalog_raw.tsv
```

15. assistant (2026-06-29T11:53:43.462Z)

```
Catalog built: 1795 files, 1.232 TB. 220 VIDEO files (~1.155 TB), 163 PROXY (.lrv, 33GB), 163
THUMB, 1249 OTHER. Note `.mp4` count is 222 but VIDEO is 220 ? the 2 difference is because 3
`._GX*.MP4` AppleDouble stubs are reclassified as OTHER... let me verify that math, but first
let me start the dedup. Stage 1: group VIDEO files by size.
```

16. assistant (2026-06-29T11:53:52.390Z)

```
#!/usr/bin/env python3
"""Stage 1: load VIDEO files from catalog_raw.tsv, group by exact byte size.
Output:
  videos.json        -> list of all VIDEO records (path,size,mtime,ext)
  size_groups.json   -> {size: [indices...]} only for groups >=2
Distinct size => unique. Same-size groups need partial-sig (stage 2).
"""
import json, os

CAT = '/home/avidullu/projects/khelsutra-guru/f_drive_index/catalog_raw.tsv'
videos = []
with open(CAT, encoding='utf-8', errors='surrogateescape') as f:
    header = f.readline()
    for line in f:
        line = line.rstrip('\n')
        parts = line.split('\t')
        if len(parts) != 6:
            continue
        path, size_s, mtime_s, mtime_iso, ext, cat = parts
        if cat != 'VIDEO':
            continue
        videos.append({
            'idx': len(videos),
            'path': path,
            'size_bytes': int(size_s),
            'mtime': float(mtime_s),
            'mtime_iso': mtime_iso,
            'ext': ext,
        })

from collections import defaultdict
by_size = defaultdict(list)
for v in videos:
    by_size[v['size_bytes']].append(v['idx'])

size_groups = {str(s): idxs for s, idxs in by_size.items() if len(idxs) >= 2}

with open('/home/avidullu/projects/khelsutra-guru/f_drive_index/videos.json', 'w') as f:
    json.dump(videos, f, indent=0)
```

```
with open('/home/avidullu/projects/khelsutra-guru/f_drive_index/size_groups.json', 'w') as f:
    json.dump(size_groups, f, indent=2)

n_unique_by_size = sum(1 for s, idxs in by_size.items() if len(idxs) == 1)
n_in_groups = sum(len(idxs) for s, idxs in by_size.items() if len(idxs) >= 2)
print('Total VIDEO files: %d' % len(videos))
print('Distinct sizes: %d' % len(by_size))
print('Sizes that are unique (1 file): %d -> these %d files are content-unique' %
(n_unique_by_size, n_unique_by_size))
print('Same-size groups (>=2 files): %d, covering %d files' % (len(size_groups), n_in_groups))
print('--- same-size groups detail ---')
for s, idxs in sorted(by_size.items(), key=lambda kv: -len(kv[1])):
    if len(idxs) >= 2:
        print('  size=%d  count=%d' % (s, len(idxs)))
        for i in idxs:
            print('      %s' % videos[i]['path'])
```

17. user (2026-06-29T11:53:53.458Z)

```
File created successfully at: /home/avidullu/projects/khelsutra-
guru/f_drive_index/dedup_stage1.py (file state is current in your context ? no need to Read it
back)
```

18. user (2026-06-29T11:53:57.353Z)

```
Total VIDEO files: 220
Distinct sizes: 175
Sizes that are unique (1 file): 134 -> these 134 files are content-unique
Same-size groups (>=2 files): 41, covering 86 files
--- same-size groups detail ---
  size=196182380  count=3
      /mnt/f/Data/Initial GoPRo with Sports Recordings/GX010053.MP4
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/2026 March
6/100GOPRO/GX010053.MP4
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Training/Golden
Labelled/MahadevpuraSingles.MP4
  size=11523369189  count=3
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 12
June/100GOPRO/Badminton/100GOPRO/GX010132.MP4
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 12
June/100GOPRO/Badminton/Vasant vs [Avi and Adarsh].MP4
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Roora June 15/Video 9 -
Badminton/GX010132.MP4
  size=10278288099  count=3
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 12
June/100GOPRO/Badminton/100GOPRO/GX020130.MP4
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 12
June/100GOPRO/Pickleball/GX020130.MP4
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Roora June 15/Video 5 -
Pickleball/GX020130.MP4
  size=11523183118  count=3
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/May 19
2026/100GOPRO/GX010104.MP4
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Roora June 15/Video 8 - Table
tennis/GX010104.MP4
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Tabletennis 10
June/GX010104.MP4
  size=662437897  count=2
      /mnt/f/Data/Initial GoPRo with Sports Recordings/GX010029.MP4
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/2026 March
6/100GOPRO/Bike/Bike going to Mayura.MP4
  size=583613613  count=2
      /mnt/f/Data/Initial GoPRo with Sports Recordings/GX010030.MP4
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/2026 March
6/100GOPRO/Bike/Bike return from Mayura.MP4
```

```
  size=128577428  count=2
      /mnt/f/Data/Initial GoPRo with Sports Recordings/GX010038.MP4
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/2026 March 6/100GOPRO/Chhetri
Opinion.MP4
  size=226442602  count=2
      /mnt/f/Data/Initial GoPRo with Sports Recordings/GX010040.MP4
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/2026 March 6/100GOPRO/ELT
Pickleball/GX010040.MP4
  size=300902350  count=2
      /mnt/f/Data/Initial GoPRo with Sports Recordings/GX010043.MP4
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/2026 March 6/100GOPRO/ELT
Pickleball/GX010043.MP4
  size=417255008  count=2
      /mnt/f/Data/Initial GoPRo with Sports Recordings/GX010044.MP4
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/2026 March 6/100GOPRO/ELT
Pickleball/GX010044.MP4
  size=155169125  count=2
      /mnt/f/Data/Initial GoPRo with Sports Recordings/GX010045.MP4
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/2026 March 6/100GOPRO/ELT
Pickleball/Pickleball recording ELT.MP4
  size=1099638878  count=2
      /mnt/f/Data/Initial GoPRo with Sports Recordings/GX010046.MP4
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/2026 March 6/100GOPRO/ELT
Pickleball/Pickleball chestie footage.MP4
  size=805247217  count=2
      /mnt/f/Data/Initial GoPRo with Sports Recordings/GX010048.MP4
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/2026 March
6/100GOPRO/Cricket/Cricket Half Pitch - 3.MP4
  size=28958174  count=2
      /mnt/f/Data/Initial GoPRo with Sports Recordings/GX010049.MP4
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/2026 March
6/100GOPRO/Cricket/Cricket Half pitch - 2.MP4
  size=1360746043  count=2
      /mnt/f/Data/Initial GoPRo with Sports Recordings/GX010050.MP4
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/2026 March
6/100GOPRO/Cricket/Cricket Half pitch.MP4
  size=794794594  count=2
      /mnt/f/Data/Initial GoPRo with Sports Recordings/GX010051.MP4
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/2026 March
6/100GOPRO/GX010051.MP4
  size=1881912767  count=2
      /mnt/f/Data/Initial GoPRo with Sports Recordings/GX010055.MP4
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Roora June 15/Video 7 -
Badminton Multicourt/GX010055.MP4
  size=1542282226  count=2
      /mnt/f/Data/Initial GoPRo with Sports Recordings/GX010056.MP4
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Misc/GX010056.MP4
  size=1243054145  count=2
      /mnt/f/Data/Initial GoPRo with Sports Recordings/GX010057.MP4
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Misc/GX010057.MP4
  size=2318282212  count=2
      /mnt/f/Data/Initial GoPRo with Sports Recordings/GX010061.MP4
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/2026 March 6/100GOPRO/Mom Jan
26 Singing.MP4
  size=11445052772  count=2
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 1 June
2026/100GOPRO/GX010108.MP4
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Training/Golden
Labelled/Adarsh and Avi.MP4
  size=9227260503  count=2
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 12 June
2026/mahadevpura-2.MP4
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Roora June 15/Video 3 -
Badminton Multicourt/mahadevpura-2.MP4
  size=11522544841  count=2
```

```
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 12
June/100GOPRO/Badminton/100GOPRO/GX010130.MP4
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 12
June/100GOPRO/Pickleball/GX010130.MP4
  size=11422240787  count=2
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 12
June/100GOPRO/Badminton/100GOPRO/GX010131.MP4
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 12
June/100GOPRO/Pickleball/GX010131.MP4
  size=5533413839  count=2
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 12
June/100GOPRO/Badminton/100GOPRO/GX010133.MP4
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 12
June/100GOPRO/Badminton/Adarsh v Avi.MP4
  size=4424901717  count=2
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 12
June/100GOPRO/Badminton/100GOPRO/GX010134.MP4
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 12
June/100GOPRO/Badminton/Adarsh v Vasant 2.MP4
  size=11522290992  count=2
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 12
June/100GOPRO/Badminton/100GOPRO/GX020132.MP4
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 12
June/100GOPRO/Badminton/All 1 v 1.MP4
  size=7294607373  count=2
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 12
June/100GOPRO/Badminton/100GOPRO/GX030132.MP4
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 12
June/100GOPRO/Badminton/Adarsh v Vasant.MP4
  size=11531882761  count=2
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 2 May
2026/100GOPRO/GX010093.MP4
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Training/Golden
Labelled/LargeTestVideo.MP4
  size=11533390543  count=2
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 2 May
2026/100GOPRO/GX020094.MP4
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Roora June 15/Video 4 -
Badminton/GX020094.MP4
  size=10611615335  count=2
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 2 May
2026/100GOPRO/GX030094.MP4
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Roora June 15/Video 6 -
Badminton/GX030094.MP4
  size=8058214679  count=2
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 22 May
2026/gBaddy/GX010106.MP4
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Training/Golden
Labelled/KushagraYashAviFirstVideo.MP4
  size=7075098663  count=2
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 22 May
2026/gBaddy/GX020107.MP4
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Training/Golden
Labelled/gBaaddy20May2026.MP4
  size=11532555534  count=2
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 4 May
2026/100GOPRO/GX010097.MP4
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Roora June 15/Video 12 -
Badminton Multicourt/Boxhill Doubles.MP4
  size=1510427943  count=2
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 4 May
2026/100GOPRO/GX030097.MP4
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Training/Golden
Labelled/TestLargeVideo.MP4
  size=11522736196  count=2
```

```
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Bdminton 15
June/100GOPRO/GX010135.MP4
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Roora June 15/Video 1 -
Badminton/GX010135.MP4
  size=4469695789  count=2
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Bdminton 15
June/100GOPRO/GX020104.MP4
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/May 19
2026/100GOPRO/GX020104.MP4
  size=10925859511  count=2
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/GoPro 20 May
2026/100GOPRO/Badminton BXH 2.MP4
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Roora June 15/Video 10 -
Badminton Multicourt/Badminton BXH 2.MP4
  size=11523637665  count=2
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/GoPro 20 May
2026/100GOPRO/Pickleball ELT 1.MP4
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Roora June 15/Video 11 -
Pickleball/Pickleball ELT 1.MP4
  size=5585880493  count=2
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/May 19
2026/100GOPRO/GX010105.MP4
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Tabletennis 10
June/GX010105.MP4
  size=11500454862  count=2
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Roora June 15/Video 2 -
Tabletennis/GX010124.MP4
      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Tabletennis 11
June/GX010124.MP4
```

19. assistant (2026-06-29T11:54:01.025Z)

```
41 same-size groups covering 86 files. Many golden clips appear as renamed copies (e.g.
`MahadevpuraSingles.MP4` == `GX010053.MP4`). Now Stage 2: compute partial signatures (md5 of
first 16MB + last 16MB) for these 86 files to confirm/split groups. Let me write that.
```

20. assistant (2026-06-29T11:54:15.035Z)

```
#!/usr/bin/env python3
"""Stage 2: for each same-size group, compute partial signature
md5(first 16MB ++ last 16MB) using direct seek reads (READ-ONLY).
Checkpoints partial sigs to partial_sigs.json (resumable).
Then split groups by (size, partial_sig); any (size,psig) group >=2 needs full md5.
"""
import json, os, hashlib, sys

BASE = '/home/avidullu/projects/khelsutra-guru/f_drive_index'
CHUNK = 16 * 1024 * 1024  # 16 MB
videos = json.load(open(f'{BASE}/videos.json'))
size_groups = json.load(open(f'{BASE}/size_groups.json'))

psig_file = f'{BASE}/partial_sigs.json'
if os.path.exists(psig_file):
    partial = json.load(open(psig_file))
else:
    partial = {}
```

indices that need a partial sig = union of all size_groups

```
need = sorted({i for idxs in size_groups.values() for i in idxs})
print('Files needing partial sig: %d' % len(need))

def partial_sig(path, size):
    h = hashlib.md5()
    with open(path, 'rb') as f:
```

```
        # first 16MB
        first = f.read(min(CHUNK, size))
        h.update(first)
        # last 16MB (if file > CHUNK; avoid double-reading small files)
        if size > CHUNK:
            tail_start = max(size - CHUNK, CHUNK)  # don't overlap with first if possible
            f.seek(size - min(CHUNK, size))
            last = f.read(CHUNK)
            h.update(last)
    return h.hexdigest()

done = 0
errors = []
for i in need:
    key = str(i)
    if key in partial:
        continue
    v = videos[i]
    try:
        ps = partial_sig(v['path'], v['size_bytes'])
        partial[key] = ps
    except Exception as e:
        errors.append((v['path'], str(e)))
        partial[key] = 'ERROR:' + str(e)[:80]
    done += 1
    if done % 10 == 0:
        json.dump(partial, open(psig_file, 'w'), indent=0)
        print('  ...%d/%d partial sigs computed' % (done, len(need)), flush=True)

json.dump(partial, open(psig_file, 'w'), indent=2)
print('Partial sigs computed this run: %d (errors=%d)' % (done, len(errors)))
for p, e in errors:
    sys.stderr.write('PSIG ERROR %s : %s\n' % (p, e))
```

Now split by (size, psig)

```
from collections import defaultdict
groups = defaultdict(list)
for s, idxs in size_groups.items():
    for i in idxs:
        ps = partial.get(str(i), 'MISSING')
        groups[(int(s), ps)].append(i)

confirm_groups = {f'{s}|{ps}': idxs for (s, ps), idxs in groups.items() if len(idxs) >= 2}
json.dump(confirm_groups, open(f'{BASE}/confirm_groups.json', 'w'), indent=2)

n_split_unique = sum(1 for k, idxs in groups.items() if len(idxs) == 1)
print('After partial-sig split:')
print('  (size,psig) groups that are now singletons (unique): %d' % n_split_unique)
print('  (size,psig) groups still >=2 (need FULL md5): %d covering %d files' % (
    len(confirm_groups), sum(len(v) for v in confirm_groups.values())))
for k, idxs in confirm_groups.items():
    print('  group %s -> %d files' % (k[:60], len(idxs)))
```

21. user (2026-06-29T11:54:16.100Z)

```
File created successfully at: /home/avidullu/projects/khelsutra-
guru/f_drive_index/dedup_stage2.py (file state is current in your context ? no need to Read it
back)
```

22. user (2026-06-29T11:54:35.443Z)

```
Files needing partial sig: 86
  ...10/86 partial sigs computed
  ...20/86 partial sigs computed
```

```
  ...30/86 partial sigs computed
  ...40/86 partial sigs computed
  ...50/86 partial sigs computed
  ...60/86 partial sigs computed
  ...70/86 partial sigs computed
  ...80/86 partial sigs computed
Partial sigs computed this run: 86 (errors=0)
After partial-sig split:
  (size,psig) groups that are now singletons (unique): 0
  (size,psig) groups still >=2 (need FULL md5): 41 covering 86 files
  group 662437897|87a62cec6a9295b963c9acbeae4c9d6b -> 2 files
  group 583613613|ee9462648bd3eacef4585d1d3077946a -> 2 files
  group 128577428|03325bc79079927f6baadd1e415d78da -> 2 files
  group 226442602|2326b031e06adf9b1aae8061bf5cd702 -> 2 files
  group 300902350|3bcedd5c711f781ececb47a9f12d71ed -> 2 files
  group 417255008|ba03428dfe5fd8ffb679684e6abd3dd7 -> 2 files
  group 155169125|05fa5bffd59c3d173515fd9d340bacef -> 2 files
  group 1099638878|ffadfe30f031b0b3d0147234d88969a0 -> 2 files
  group 805247217|4604142d3267f7089868f521c3881608 -> 2 files
  group 28958174|b7773404879b34f84d272d020fd959a2 -> 2 files
  group 1360746043|3b10d64f0bb7318516883c729f0b6053 -> 2 files
  group 794794594|9368b3a3fcb0cb26ab1a44e99e62ea29 -> 2 files
  group 196182380|fc364b1dc8927bad694c5c48fa227117 -> 3 files
  group 1881912767|211187e0496bdcf382757dc8326401a4 -> 2 files
  group 1542282226|6e7c2ffa3d991125863c4279d83a49c9 -> 2 files
  group 1243054145|59ed13a5c72d0b8b3692d177e306413a -> 2 files
  group 2318282212|21e29f44b0e38f676a4435ac50b67983 -> 2 files
  group 11445052772|b782f3cf539c900913cca978d49f0138 -> 2 files
  group 9227260503|12b764f50297b45f9f38c1560093cad0 -> 2 files
  group 11522544841|069b7ec587d58b090a9dab4ebf32a16f -> 2 files
  group 11422240787|f6dbe33b53d1c372b13653df3d616bd1 -> 2 files
  group 11523369189|285518e0e723e22b0f74f235f0dde790 -> 3 files
  group 5533413839|2076c82f80b59a186e4a17ec0f2b26e8 -> 2 files
  group 4424901717|95d05121b689a7b484ed6d8bdf64e06d -> 2 files
  group 10278288099|75fa0270625e4e3d05ca8265530901c9 -> 3 files
  group 11522290992|dc0163af8a61cbae434defe8841c50a7 -> 2 files
  group 7294607373|05486280ac7d6d072ef351f5058f2daf -> 2 files
  group 11531882761|3a3f66c7344fc2b21ca252bb0ef0db0c -> 2 files
  group 11533390543|834ed20fe48ac1f2e86efd08234155a8 -> 2 files
  group 10611615335|aa8726cb6a306c8939a2cc4e2c7ae8f1 -> 2 files
  group 8058214679|ef10c8948ebb4e67acbaebf4735c04ea -> 2 files
  group 7075098663|e2aa038d349658bffee654113755e1a1 -> 2 files
  group 11532555534|f19b28760a9f4c8b7daf51f1a5770dd5 -> 2 files
  group 1510427943|6f48ba6f71537a8a63d7b1f41957923d -> 2 files
  group 11522736196|2dcaae6a57a0bd851c409544ecf5c3be -> 2 files
  group 4469695789|23a02c319878fd2b70da7c2d07070585 -> 2 files
  group 10925859511|c63c268d54c435b08a8a0b4cdb064349 -> 2 files
  group 11523637665|34df329cf8b56800f03d18e884d097e8 -> 2 files
  group 11523183118|8a58729403b9607d74b25e0431124044 -> 3 files
  group 5585880493|579367c03ae827a9c9d2baf4e4a9cc7c -> 2 files
  group 11500454862|1b03816eca0fcc896a35d0a55a7c2bdc -> 2 files
--- stage2 stderr ---
```

23. assistant (2026-06-29T11:54:39.366Z)

All 41 groups survived the partial-sig check (same size AND same head+tail signature) ? these are very likely true duplicates. Stage 3: full MD5 to confirm. This is the heavy step (86 files, many ~11GB), so I'll checkpoint per-file. Let me compute full MD5 for these 86 files. I'll run it in the background since it involves reading ~600GB.

24. assistant (2026-06-29T11:54:57.351Z)

```
#!/usr/bin/env python3
"""Compute full-file MD5 (= video_id, per backend/utils/hashing.py::compute_video_id)
for a given set of video indices. Checkpoints to full_md5.json after EACH file so
```

```
the run is resumable and partial results survive. READ-ONLY.

Usage: fullmd5.py <mode>
  mode=confirm  -> only the files in confirm_groups (dup candidates) + golden-name matches
  mode=all      -> all VIDEO files not yet hashed (progressive, for remaining uniques)
"""
import json, os, hashlib, sys, time

BASE = '/home/avidullu/projects/khelsutra-guru/f_drive_index'
videos = json.load(open(f'{BASE}/videos.json'))
mode = sys.argv[1] if len(sys.argv) > 1 else 'confirm'

GOLDEN = {'adarsh_avi_singles','kushagra_singles','largetest_doubles','gbaaddy',
 'testlarge_short','mahadevpura_singles','mahadevpura_2','gx010128','kushagra_singles',
'mahadevpura_1','gx030094','badminton_bxh_2','gx010141','gx010137','boxhill_doubles','gx020094'}

md5_file = f'{BASE}/full_md5.json'
if os.path.exists(md5_file):
    md5map = json.load(open(md5_file))
else:
    md5map = {}
```

Determine target indices

```
targets = set()
if mode == 'confirm':
    cg = json.load(open(f'{BASE}/confirm_groups.json'))
    for idxs in cg.values():
        targets.update(idxs)
    # plus golden-name basename matches (loose normalization)
    for v in videos:
        b = os.path.splitext(os.path.basename(v['path']))[0].lower()
        norm = b.replace(' ', '').replace('-', '').replace('_', '')
        for g in GOLDEN:
            gn = g.replace('_', '').replace('-', '')
            if gn in norm or norm in gn:
                targets.add(v['idx'])
                break
elif mode == 'all':
    targets = set(range(len(videos)))
else:
    print('unknown mode'); sys.exit(1)

todo = [i for i in sorted(targets) if str(i) not in md5map]
print('mode=%s targets=%d already_hashed=%d todo=%d' % (
    mode, len(targets), len(targets)-len(todo), len(todo)), flush=True)

def md5_full(path):
    h = hashlib.md5()
    with open(path, 'rb') as f:
        for block in iter(lambda: f.read(8*1024*1024), b''):
            h.update(block)
    return h.hexdigest()

errlog = open(f'{BASE}/progress.log', 'a')
done = 0
t0 = time.time()
for i in todo:
    v = videos[i]
    st = time.time()
    try:
        digest = md5_full(v['path'])
        md5map[str(i)] = digest
    except Exception as e:
        md5map[str(i)] = 'ERROR:' + str(e)[:100]
```

```
        errlog.write('[%s] FULLMD5 ERROR %s : %s\n' % (time.strftime('%Y-%m-%dT%H:%M:%S'),
v['path'], e))
        errlog.flush()
    # checkpoint after EACH file
    json.dump(md5map, open(md5_file, 'w'), indent=0)
    done += 1
    dt = time.time() - st
    gb = v['size_bytes']/1e9
    print('  [%d/%d] %.2fGB in %.1fs (%.0fMB/s) %s' % (
        done, len(todo), gb, dt, gb*1000/max(dt,0.001), os.path.basename(v['path'])),
flush=True)
    errlog.write('[%s] hashed [%d/%d] %s -> %s\n' % (
        time.strftime('%Y-%m-%dT%H:%M:%S'), done, len(todo), v['path'], md5map[str(i)][:32]))
    errlog.flush()

print('DONE mode=%s hashed_this_run=%d total_elapsed=%.0fs' % (mode, done, time.time()-t0),
flush=True)
errlog.write('[%s] fullmd5 mode=%s DONE hashed=%d elapsed=%.0fs\n' % (
    time.strftime('%Y-%m-%dT%H:%M:%S'), mode, done, time.time()-t0))
errlog.close()
```

25. user (2026-06-29T11:54:58.428Z)

```
File created successfully at: /home/avidullu/projects/khelsutra-guru/f_drive_index/fullmd5.py
(file state is current in your context ? no need to Read it back)
```

26. assistant (2026-06-29T11:55:01.569Z)

```
Now let me kick off the confirm-mode full MD5 in the background (it reads ~600GB across the 86
dup-candidate + golden files), and meanwhile work on ffprobe metadata and GCS cross-reference in
parallel.
```

27. user (2026-06-29T11:55:05.765Z)

```
Command running in background with ID: b7gghqsye. Output is being written to:
/tmp/claude-1000/-home-avidullu-projects-khelsutra-
guru/4f364eb7-a600-4a2c-9b37-e5a313b6db9d/tasks/b7gghqsye.output. You will be notified when it
completes. To check interim output, use Read on that file path.
```

28. assistant (2026-06-29T11:55:08.138Z)

```
While that runs, let me start the ffprobe metadata pass (header-only, fast) and the GCS cross-
reference in parallel.
```

29. assistant (2026-06-29T11:55:18.287Z)

```
#!/usr/bin/env python3
"""Header-only ffprobe metadata per VIDEO file (fast; no frame decode).
Captures duration_s, fps, width, height, codec. Checkpoints to ffprobe_meta.json.
READ-ONLY.
"""
import json, os, subprocess, sys, time

BASE = '/home/avidullu/projects/khelsutra-guru/f_drive_index'
videos = json.load(open(f'{BASE}/videos.json'))
out_file = f'{BASE}/ffprobe_meta.json'
meta = json.load(open(out_file)) if os.path.exists(out_file) else {}

def probe(path):
    cmd = ['ffprobe', '-v', 'error', '-select_streams', 'v:0',
           '-show_entries',
'stream=codec_name,width,height,r_frame_rate,avg_frame_rate:format=duration',
           '-of', 'json', path]
    p = subprocess.run(cmd, capture_output=True, text=True, timeout=60)
```

```
    if p.returncode != 0:
        return {'error': p.stderr.strip()[:120]}
    j = json.loads(p.stdout)
    st = (j.get('streams') or [{}])[0]
    fmt = j.get('format') or {}
    def parse_rate(r):
        try:
            n, d = r.split('/')
            d = float(d)
            return round(float(n)/d, 3) if d else None
        except Exception:
            return None
    fps = parse_rate(st.get('avg_frame_rate') or '') or parse_rate(st.get('r_frame_rate') or '')
    dur = fmt.get('duration')
    return {
        'codec': st.get('codec_name'),
        'width': st.get('width'),
        'height': st.get('height'),
        'fps': fps,
        'duration_s': round(float(dur), 3) if dur else None,
    }

todo = [v for v in videos if str(v['idx']) not in meta]
print('ffprobe todo=%d already=%d' % (len(todo), len(meta)), flush=True)
log = open(f'{BASE}/progress.log', 'a')
done = 0
for v in todo:
    try:
        meta[str(v['idx'])] = probe(v['path'])
    except Exception as e:
        meta[str(v['idx'])] = {'error': str(e)[:120]}
        log.write('[%s] FFPROBE ERROR %s : %s\n' % (time.strftime('%H:%M:%S'), v['path'], e))
    done += 1
    if done % 20 == 0:
        json.dump(meta, open(out_file, 'w'), indent=0)
        print('  ...%d/%d' % (done, len(todo)), flush=True)
json.dump(meta, open(out_file, 'w'), indent=2)
log.write('[%s] ffprobe done, probed=%d\n' % (time.strftime('%H:%M:%S'), done))
log.close()
print('ffprobe DONE probed=%d' % done)
```

30. user (2026-06-29T11:55:19.361Z)

```
File created successfully at: /home/avidullu/projects/khelsutra-
guru/f_drive_index/ffprobe_meta.py (file state is current in your context ? no need to Read it
back)
```

31. user (2026-06-29T11:55:39.649Z)

```
ffprobe todo=220 already=0
  ...20/220
  ...40/220
  ...60/220
  ...80/220
  ...100/220
  ...120/220
  ...140/220
  ...160/220
  ...180/220
  ...200/220
  ...220/220
ffprobe DONE probed=220
```

32. assistant (2026-06-29T11:55:42.179Z)

ffprobe metadata complete for all 220 videos. Now the GCS cross-reference. Let me list the corpus bucket.

33. user (2026-06-29T11:56:04.789Z)

```
=== gs://khelsutra-rally-corpus/videos/ ===
 767766440  2026-06-22T13:42:54Z  gs://khelsutra-rally-corpus/videos/GX010142_1080p.mp4
 632948235  2026-06-22T13:42:46Z  gs://khelsutra-rally-corpus/videos/GX010143_1080p.mp4
 659514891  2026-06-22T13:42:48Z  gs://khelsutra-rally-corpus/videos/GX010147_1080p.mp4
1429944539  2026-06-20T14:27:35Z  gs://khelsutra-rally-corpus/videos/GX030094_proxy.mp4
 548827049  2026-06-20T14:26:23Z  gs://khelsutra-rally-corpus/videos/mahadevpura-1.MP4
 548827049  2026-06-22T10:21:57Z  gs://khelsutra-rally-corpus/videos/mahadevpura_1.mp4
  90194321  2026-06-20T14:20:32Z  gs://khelsutra-rally-corpus/videos/mahadevpura_smoke_180s.mp4
                                  gs://khelsutra-rally-corpus/videos/SHANAN/
TOTAL: 7 objects, 4678022524 bytes (4.36GiB)
=== EXIT 0 ===
```

34. assistant (2026-06-29T11:56:06.912Z)

The bucket has only 7 objects in `videos/` (plus a SHANAN subfolder). Let me also list the whole bucket recursively and the SHANAN subfolder to be thorough.

35. user (2026-06-29T11:56:31.877Z)

```
=== recursive bucket listing ===
gs://khelsutra-rally-corpus/:

gs://khelsutra-rally-corpus/archive/:

gs://khelsutra-rally-corpus/archive/droplet-claude-do-rally-test-20260625/:
     12288  2026-06-25T11:15:03Z  gs://khelsutra-rally-corpus/archive/droplet-claude-do-rally-
test-20260625/access.db
       368  2026-06-25T11:15:03Z  gs://khelsutra-rally-corpus/archive/droplet-claude-do-rally-
test-20260625/engine-config.json

gs://khelsutra-rally-corpus/archive/droplet-claude-do-rally-test-20260625/output/:
  98273085  2026-06-25T11:15:16Z  gs://khelsutra-rally-corpus/archive/droplet-claude-do-rally-
test-20260625/output/highlights-one.mp4
  25284972  2026-06-25T11:15:10Z  gs://khelsutra-rally-corpus/archive/droplet-claude-do-rally-
test-20260625/output/highlights-shannon.mp4
  28019156  2026-06-25T11:15:08Z  gs://khelsutra-rally-corpus/archive/droplet-claude-do-rally-
test-20260625/output/highlights.mp4
    135168  2026-06-25T11:15:04Z  gs://khelsutra-rally-corpus/archive/droplet-claude-do-rally-
test-20260625/output/sports_indexer.db
         5  2026-06-25T11:15:03Z  gs://khelsutra-rally-corpus/archive/droplet-claude-do-rally-
test-20260625/output/sports_indexer.db.lock

gs://khelsutra-rally-corpus/archive/droplet-claude-do-rally-test-20260625/uploads/:
  58145391  2026-06-25T11:15:13Z  gs://khelsutra-rally-corpus/archive/droplet-claude-do-rally-
test-20260625/uploads/ShannanShortClip.mp4
 125410307  2026-06-25T11:15:18Z  gs://khelsutra-rally-corpus/archive/droplet-claude-do-rally-
test-20260625/uploads/SmokeTestBadmintonClip.mp4
    175807  2026-06-25T11:15:04Z  gs://khelsutra-rally-corpus/archive/droplet-claude-do-rally-
test-20260625/uploads/sample.mp4

gs://khelsutra-rally-corpus/highlights/:
  22383681  2026-06-20T17:46:35Z  gs://khelsutra-rally-
corpus/highlights/curated_top10_longest.mp4
     16594  2026-06-21T22:24:34Z  gs://khelsutra-rally-corpus/highlights/p3_app.log
  77309429  2026-06-21T22:24:30Z  gs://khelsutra-rally-
corpus/highlights/p3_packaging_highlights.mp4
     57279  2026-06-21T22:24:32Z  gs://khelsutra-rally-corpus/highlights/p3_run.log

gs://khelsutra-rally-corpus/labels/:
```

```
      4106  2026-06-22T10:01:41Z  gs://khelsutra-rally-
corpus/labels/2026-06-08_adarsh_avi_singles.rallies.csv
      2049  2026-06-22T10:02:13Z  gs://khelsutra-rally-
corpus/labels/2026-06-08_gbaaddy.rallies.csv
      1413  2026-06-22T10:01:52Z  gs://khelsutra-rally-
corpus/labels/2026-06-08_kushagra_singles.rallies.csv
      2140  2026-06-22T10:02:02Z  gs://khelsutra-rally-
corpus/labels/2026-06-08_largetest_doubles.rallies.csv
       287  2026-06-22T10:02:24Z  gs://khelsutra-rally-
corpus/labels/2026-06-08_testlarge_short.rallies.csv
      1325  2026-06-22T10:02:35Z  gs://khelsutra-rally-
corpus/labels/2026-06-09_mahadevpura_singles.rallies.csv
      2286  2026-06-22T10:02:56Z  gs://khelsutra-rally-
corpus/labels/2026-06-16_GX010128.rallies.csv
      1719  2026-06-22T10:02:45Z  gs://khelsutra-rally-
corpus/labels/2026-06-16_mahadevpura_2.rallies.csv
      1970  2026-06-22T10:03:31Z  gs://khelsutra-rally-
corpus/labels/2026-06-17_Badminton_BXH_2.rallies.csv
      1800  2026-06-22T10:03:17Z  gs://khelsutra-rally-
corpus/labels/2026-06-17_GX030094.rallies.csv
       465  2026-06-22T10:03:07Z  gs://khelsutra-rally-
corpus/labels/2026-06-17_mahadevpura_1.rallies.csv
      1915  2026-06-22T10:04:03Z  gs://khelsutra-rally-
corpus/labels/2026-06-18_Boxhill_Doubles.rallies.csv
      1489  2026-06-22T10:03:52Z  gs://khelsutra-rally-
corpus/labels/2026-06-18_GX010137.rallies.csv
      1274  2026-06-22T10:03:41Z  gs://khelsutra-rally-
corpus/labels/2026-06-18_GX010141.rallies.csv
      1715  2026-06-22T10:04:14Z  gs://khelsutra-rally-
corpus/labels/2026-06-18_GX020094.rallies.csv

gs://khelsutra-rally-corpus/manifests/:
      3840  2026-06-22T10:04:19Z  gs://khelsutra-rally-corpus/manifests/golden.json

gs://khelsutra-rally-corpus/models/:
   6102633  2026-06-20T14:19:50Z  gs://khelsutra-rally-corpus/models/wasb_badminton_best.pth.tar

gs://khelsutra-rally-corpus/outputs/:

gs://khelsutra-rally-corpus/outputs/05e0e577029daafcf425ca70c0c4f0a3/:
  31749944  2026-06-20T15:12:37Z  gs://khelsutra-rally-
corpus/outputs/05e0e577029daafcf425ca70c0c4f0a3/highlight_smoke_l4.mp4
       427  2026-06-21T13:29:41Z  gs://khelsutra-rally-
corpus/outputs/05e0e577029daafcf425ca70c0c4f0a3/intervals.csv
      1760  2026-06-21T13:29:41Z  gs://khelsutra-rally-
corpus/outputs/05e0e577029daafcf425ca70c0c4f0a3/report.json

gs://khelsutra-rally-corpus/outputs/parity/:
     38457  2026-06-20T14:48:03Z  gs://khelsutra-rally-corpus/outputs/parity/native_smoke_l4.csv

gs://khelsutra-rally-corpus/promo/:
    776632  2026-06-20T18:16:41Z  gs://khelsutra-rally-corpus/promo/full_cuj_real.mp4
   1670821  2026-06-21T13:28:04Z  gs://khelsutra-rally-corpus/promo/khelsutra-bucket-cuj-
screencast.mp4
    342162  2026-06-21T07:23:28Z  gs://khelsutra-rally-
corpus/promo/khelsutra-i18n-screencast.mp4
   4082564  2026-06-20T17:32:49Z  gs://khelsutra-rally-corpus/promo/promo_cuj.gif
    709229  2026-06-20T17:32:49Z  gs://khelsutra-rally-corpus/promo/promo_cuj.mp4

gs://khelsutra-rally-corpus/trajectories/:
   1549493  2026-06-22T10:03:24Z  gs://khelsutra-rally-corpus/trajectories/Badminton_BXH_2.csv
    838358  2026-06-22T10:03:58Z  gs://khelsutra-rally-corpus/trajectories/Boxhill_Doubles.csv
   1543847  2026-06-22T10:02:50Z  gs://khelsutra-rally-corpus/trajectories/GX010128.csv
    797173  2026-06-22T10:03:47Z  gs://khelsutra-rally-corpus/trajectories/GX010137.csv
    776018  2026-06-22T10:03:36Z  gs://khelsutra-rally-corpus/trajectories/GX010141.csv
```

```
    608149  2026-06-22T10:04:09Z  gs://khelsutra-rally-corpus/trajectories/GX020094.csv
    701986  2026-06-22T10:03:12Z  gs://khelsutra-rally-corpus/trajectories/GX030094.csv
   2476375  2026-06-22T10:01:36Z  gs://khelsutra-rally-
corpus/trajectories/adarsh_avi_singles.csv
   1702622  2026-06-22T10:02:08Z  gs://khelsutra-rally-corpus/trajectories/gbaaddy.csv
   1159102  2026-06-22T10:01:46Z  gs://khelsutra-rally-corpus/trajectories/kushagra_singles.csv
    693876  2026-06-22T10:01:57Z  gs://khelsutra-rally-corpus/trajectories/largetest_doubles.csv
    459324  2026-06-22T10:03:01Z  gs://khelsutra-rally-corpus/trajectories/mahadevpura_1.csv
   1454557  2026-06-22T10:02:40Z  gs://khelsutra-rally-corpus/trajectories/mahadevpura_2.csv
=== EXIT 0 ===
```

36. assistant (2026-06-29T11:56:34.994Z)

Now I have a clear picture. The bucket has labels/trajectories for all 15 golden clips, but `videos/` only has a handful of actual video objects. Let me see the rest of the listing (SHANAN subfolder) and the golden manifest to map golden names to source files.

37. user (2026-06-29T11:57:00.799Z)

```
=== tail of full ls (SHANAN etc) ===
gs://khelsutra-rally-corpus/videos/SHANAN/:
 691158901  2026-06-23T13:42:59Z  gs://khelsutra-rally-corpus/videos/SHANAN/VID_1.mp4
2794800788  2026-06-23T13:43:14Z  gs://khelsutra-rally-corpus/videos/SHANAN/VID_2.mp4
2627544458  2026-06-23T13:43:26Z  gs://khelsutra-rally-corpus/videos/SHANAN/VID_3.mp4

gs://khelsutra-rally-corpus/weights/:

gs://khelsutra-rally-corpus/weights/0.1.0-l4/:
      3028  2026-06-20T17:59:14Z  gs://khelsutra-rally-corpus/weights/0.1.0-l4/manifest.json
       836  2026-06-20T17:59:15Z  gs://khelsutra-rally-corpus/weights/0.1.0-l4/weights.json
TOTAL: 67 objects, 11287534406 bytes (10.51GiB)
=== all objects under videos/ (from full ls) ===
 767766440  2026-06-22T13:42:54Z  gs://khelsutra-rally-corpus/videos/GX010142_1080p.mp4
 632948235  2026-06-22T13:42:46Z  gs://khelsutra-rally-corpus/videos/GX010143_1080p.mp4
 659514891  2026-06-22T13:42:48Z  gs://khelsutra-rally-corpus/videos/GX010147_1080p.mp4
1429944539  2026-06-20T14:27:35Z  gs://khelsutra-rally-corpus/videos/GX030094_proxy.mp4
 548827049  2026-06-20T14:26:23Z  gs://khelsutra-rally-corpus/videos/mahadevpura-1.MP4
 548827049  2026-06-22T10:21:57Z  gs://khelsutra-rally-corpus/videos/mahadevpura_1.mp4
  90194321  2026-06-20T14:20:32Z  gs://khelsutra-rally-corpus/videos/mahadevpura_smoke_180s.mp4
 691158901  2026-06-23T13:42:59Z  gs://khelsutra-rally-corpus/videos/SHANAN/VID_1.mp4
2794800788  2026-06-23T13:43:14Z  gs://khelsutra-rally-corpus/videos/SHANAN/VID_2.mp4
2627544458  2026-06-23T13:43:26Z  gs://khelsutra-rally-corpus/videos/SHANAN/VID_3.mp4
=== golden manifest ===
```

```
[
  {
    "name": "adarsh_avi_singles",
    "trajectory": "gs://khelsutra-rally-corpus/trajectories/adarsh_avi_singles.csv",
    "fps": 59.94,
    "frame_width": 1920,
    "labels": "gs://khelsutra-rally-corpus/labels/2026-06-08_adarsh_avi_singles.rallies.csv"
  },
  {
    "name": "kushagra_singles",
    "trajectory": "gs://khelsutra-rally-corpus/trajectories/kushagra_singles.csv",
    "fps": 59.94,
    "frame_width": 1920,
    "labels": "gs://khelsutra-rally-corpus/labels/2026-06-08_kushagra_singles.rallies.csv"
  },
  {
    "name": "largetest_doubles",
    "trajectory": "gs://khelsutra-rally-corpus/trajectories/largetest_doubles.csv",
    "fps": 29.97,
    "frame_width": 1920,
    "labels": "gs://khelsutra-rally-corpus/labels/2026-06-08_largetest_doubles.rallies.csv"
  },
```

```
{
  "name": "gbaaddy",
  "trajectory": "gs://khelsutra-rally-corpus/trajectories/gbaaddy.csv",
  "fps": 59.94,
  "frame_width": 1920,
  "labels": "gs://khelsutra-rally-corpus/labels/2026-06-08_gbaaddy.rallies.csv"
},
{
  "name": "testlarge_short",
  "trajectory": "gs://khelsutra-rally-corpus/trajectories/testlarge_short.csv",
  "fps": 29.97,
  "frame_width": 1920,
  "labels": "gs://khelsutra-rally-corpus/labels/2026-06-08_testlarge_short.rallies.csv"
},
{
  "name": "mahadevpura_singles",
  "trajectory": "gs://khelsutra-rally-corpus/trajectories/mahadevpura_singles.csv",
  "fps": 29.97,
  "frame_width": 1920,
  "labels": "gs://khelsutra-rally-corpus/labels/2026-06-09_mahadevpura_singles.rallies.csv"
},
{
  "name": "mahadevpura_2",
  "trajectory": "gs://khelsutra-rally-corpus/trajectories/mahadevpura_2.csv",
  "fps": 59.94,
  "frame_width": 1920.0,
  "labels": "gs://khelsutra-rally-corpus/labels/2026-06-16_mahadevpura_2.rallies.csv"
},
{
  "name": "GX010128",
  "trajectory": "gs://khelsutra-rally-corpus/trajectories/GX010128.csv",
  "fps": 59.94,
  "frame_width": 1920.0,
  "labels": "gs://khelsutra-rally-corpus/labels/2026-06-16_GX010128.rallies.csv"
},
{
  "name": "mahadevpura_1",
  "trajectory": "gs://khelsutra-rally-corpus/trajectories/mahadevpura_1.csv",
  "fps": 59.94,
  "frame_width": 1920.0,
  "labels": "gs://khelsutra-rally-corpus/labels/2026-06-17_mahadevpura_1.rallies.csv"
},
{
  "name": "GX030094",
  "trajectory": "gs://khelsutra-rally-corpus/trajectories/GX030094.csv",
  "fps": 29.97,
  "frame_width": 1920.0,
  "labels": "gs://khelsutra-rally-corpus/labels/2026-06-17_GX030094.rallies.csv"
},
{
  "name": "Badminton_BXH_2",
  "trajectory": "gs://khelsutra-rally-corpus/trajectories/Badminton_BXH_2.csv",
  "fps": 59.94,
  "frame_width": 1920.0,
  "labels": "gs://khelsutra-rally-corpus/labels/2026-06-17_Badminton_BXH_2.rallies.csv"
},
{
  "name": "GX010141",
  "trajectory": "gs://khelsutra-rally-corpus/trajectories/GX010141.csv",
  "fps": 59.94,
  "frame_width": 1920.0,
  "labels": "gs://khelsutra-rally-corpus/labels/2026-06-18_GX010141.rallies.csv"
},
{
  "name": "GX010137",
```

```
    "trajectory": "gs://khelsutra-rally-corpus/trajectories/GX010137.csv",
    "fps": 59.94,
    "frame_width": 1920.0,
    "labels": "gs://khelsutra-rally-corpus/labels/2026-06-18_GX010137.rallies.csv"
  },
  {
    "name": "Boxhill_Doubles",
    "trajectory": "gs://khelsutra-rally-corpus/trajectories/Boxhill_Doubles.csv",
    "fps": 29.97,
    "frame_width": 1920.0,
    "labels": "gs://khelsutra-rally-corpus/labels/2026-06-18_Boxhill_Doubles.rallies.csv"
  },
  {
    "name": "GX020094",
    "trajectory": "gs://khelsutra-rally-corpus/trajectories/GX020094.csv",
    "fps": 29.97,
    "frame_width": 1920.0,
    "labels": "gs://khelsutra-rally-corpus/labels/2026-06-18_GX020094.rallies.csv"
  }
]
```


38. assistant (2026-06-29T11:57:04.795Z)

This confirms the DATA_IN_GCS gap: the corpus has trajectories+labels for all 15 golden clips, but `videos/` only contains a few actual source videos. The full-res golden source videos are NOT in the bucket (only `mahadevpura_1` exists as full-res, plus a `GX030094_proxy`, plus `mahadevpura_smoke_180s` which is a derived smoke clip).

Let me check on the background full-MD5 job, then build the golden-name ? F: path mapping.

39. user (2026-06-29T11:57:09.845Z)

```
=== fullmd5 confirm progress ===
  [26/94] 0.42GB in 2.0s (208MB/s) GX010044.MP4
  [27/94] 1.10GB in 5.4s (204MB/s) Pickleball chestie footage.MP4
  [28/94] 0.16GB in 0.8s (197MB/s) Pickleball recording ELT.MP4
  [29/94] 0.79GB in 4.2s (187MB/s) GX010051.MP4
  [30/94] 0.20GB in 1.1s (178MB/s) GX010053.MP4
  [31/94] 2.32GB in 12.9s (179MB/s) Mom Jan 26 Singing.MP4
=== files hashed so far ===
hashed: 31
```

40. assistant (2026-06-29T11:57:13.427Z)

Hashing at ~190 MB/s, 31/94 done. The confirm set is 94 files (86 dup-candidates + 8 extra golden-name matches). It will read several hundred GB, so this takes a while. Let me map golden names to F: paths while it runs. Let me build the golden mapping based on the directory structure I've seen ? there's a `Training/Golden Labelled/` folder with renamed copies.

41. user (2026-06-29T11:57:19.896Z)

```
=== Golden Labelled folder ===
total 38905856
drwxrwxrwx 1 root root         512 Jun 10 12:08 .
drwxrwxrwx 1 root root         512 Jun 13 14:10 ..
-rwxrwxrwx 1 root root 11445052772 Jun  1 03:08 Adarsh and Avi.MP4
-rwxrwxrwx 1 root root        4106 Jun  8 11:50 Adarsh and Avi.rallies.csv
-rwxrwxrwx 1 root root        1970 Jun 17 08:09 Badminton_BXH_2.rallies.csv
-rwxrwxrwx 1 root root        1915 Jun 18 06:19 Boxhill_Doubles.rallies.csv
-rwxrwxrwx 1 root root        2286 Jun 16 16:57 GX010128.rallies.csv
-rwxrwxrwx 1 root root        1715 Jun 18 06:46 GX020094.rallies.csv
-rwxrwxrwx 1 root root        1800 Jun 17 07:45 GX030094.rallies.csv
-rwxrwxrwx 1 root root  8058214679 May 22 02:32 KushagraYashAviFirstVideo.MP4
-rwxrwxrwx 1 root root        1413 Jun  8 12:11 KushagraYashAviFirstVideo.rallies.csv
```

```
-rwxrwxrwx 1 root root        3271 Jun 10 12:08 KushagraYashAviFirstVideo.report.json
-rwxrwxrwx 1 root root        1611 Jun 10 12:08 KushagraYashAviFirstVideo.report.md
-rwxrwxrwx 1 root root         493 Jun 10 12:08 KushagraYashAviFirstVideo.summary.md
-rwxrwxrwx 1 root root 11531882761 May  4 11:48 LargeTestVideo.MP4
-rwxrwxrwx 1 root root        2140 Jun  8 14:58 LargeTestVideo.rallies.csv
-rwxrwxrwx 1 root root   196182380 Jan 16 02:30 MahadevpuraSingles.MP4
-rwxrwxrwx 1 root root        1325 Jun  9 02:38 MahadevpuraSingles.rallies.csv
-rwxrwxrwx 1 root root        3222 Jun 10 10:59 MahadevpuraSingles.report.json
-rwxrwxrwx 1 root root        1571 Jun 10 10:59 MahadevpuraSingles.report.md
-rwxrwxrwx 1 root root         486 Jun 10 10:59 MahadevpuraSingles.summary.md
-rwxrwxrwx 1 root root  1510427943 May  4 15:09 TestLargeVideo.MP4
-rwxrwxrwx 1 root root         287 Jun  8 16:03 TestLargeVideo.rallies.csv
-rwxrwxrwx 1 root root  7075098663 May 22 03:10 gBaaddy20May2026.MP4
-rwxrwxrwx 1 root root        2049 Jun  8 16:23 gBaaddy20May2026.rallies.csv
-rwxrwxrwx 1 root root         465 Jun 17 07:19 mahadevpura_1.rallies.csv
-rwxrwxrwx 1 root root        1719 Jun 16 14:01 mahadevpura_2.rallies.csv
=== Training folder top ===
total 0
drwxrwxrwx 1 root root 512 Jun 13 14:10 .
drwxrwxrwx 1 root root 512 Jun  7 06:16 ..
drwxrwxrwx 1 root root 512 Jun 10 12:08 Golden Labelled
=== search for golden basenames among VIDEO paths in catalog ===
```

adarsh_avi ###

kushagra ###

```
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Highlights Generator
Data/FinalOutputOfKushagraYashAviFirstVideo.mp4
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Training/Golden
Labelled/KushagraYashAviFirstVideo.MP4
/mnt/f/[Khelsutra] GoPro Hero Black 12/KhelSutra_work/kushagra_june12_proxies/GX010128.MP4
/mnt/f/[Khelsutra] GoPro Hero Black 12/KhelSutra_work/kushagra_june12_proxies/GX010129.MP4
/mnt/f/[Khelsutra] GoPro Hero Black 12/KhelSutra_work/kushagra_june12_proxies/GX020125.MP4
/mnt/f/[Khelsutra] GoPro Hero Black 12/KhelSutra_work/kushagra_june12_proxies/mahadevpura-0.MP4
/mnt/f/[Khelsutra] GoPro Hero Black 12/KhelSutra_work/kushagra_june12_proxies/mahadevpura-1.MP4
/mnt/f/[Khelsutra] GoPro Hero Black 12/KhelSutra_work/kushagra_june12_proxies/mahadevpura-2.MP4
```

largetest ###

```
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Highlights Generator
Data/FinalOutputOfVeryLargeTestVideo_GeminiGen2Prompt.mp4
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Highlights Generator
Data/Tests/Compressed_VeryLargeTestVideo.mp4
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Highlights Generator
Data/Tests/FinalOutputOfVeryLargeTestVideo_CoachPrompt.mp4
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Highlights Generator
Data/Tests/OutputOfVeryLargeTestVideo.mp4
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Highlights Generator
Data/Tests/OutputOfVeryLargeTestVideo2.mp4
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Training/Golden
Labelled/LargeTestVideo.MP4
```

gbaaddy ###

```
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Training/Golden
Labelled/gBaaddy20May2026.MP4
```

gbaddy ###

```
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 22 May
2026/gBaddy/GX010106.MP4
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 22 May
2026/gBaddy/GX010107.MP4
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 22 May
2026/gBaddy/GX020107.MP4
```

testlarge ###

```
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Highlights Generator
Data/Tests/FinalOutputOfTestLargeVideo.mp4
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Highlights Generator
Data/Tests/FinalOutputOfTestLargeVideo_GeminiGen2Prompt.mp4
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Highlights Generator
Data/Tests/FinalOutputOfTestLargeVideo_GeminiGenPrompt.mp4
```

/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Highlights Generator Data/Tests/FirstOutputOfTestLargeVideo.mp4
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Highlights Generator Data/Tests/OutputOfTestLargeVideo_CoachPrompt.mp4
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Training/Golden Labelled/TestLargeVideo.MP4

mahadevpura ###

/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 12 June 2026/mahadevpura-0.MP4
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 12 June 2026/mahadevpura-1.MP4
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 12 June 2026/mahadevpura-2.MP4
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Roora June 15/Video 3 - Badminton Multicourt/mahadevpura-2.MP4
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Training/Golden Labelled/MahadevpuraSingles.MP4
/mnt/f/[Khelsutra] GoPro Hero Black 12/KhelSutra_work/kushagra_june12_proxies/mahadevpura-0.MP4
/mnt/f/[Khelsutra] GoPro Hero Black 12/KhelSutra_work/kushagra_june12_proxies/mahadevpura-1.MP4
/mnt/f/[Khelsutra] GoPro Hero Black 12/KhelSutra_work/kushagra_june12_proxies/mahadevpura-2.MP4

GX010128 ###

/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 12 June 2026/GX010128.MP4
/mnt/f/[Khelsutra] GoPro Hero Black 12/KhelSutra_work/kushagra_june12_proxies/GX010128.MP4

GX030094 ###

/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 2 May 2026/100GOPRO/GX030094.MP4
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Roora June 15/Video 6 - Badminton/GX030094.MP4

Badminton.BXH ###

/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/GoPro 20 May 2026/100GOPRO/Badminton BXH 1.MP4
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/GoPro 20 May 2026/100GOPRO/Badminton BXH 2.MP4
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/GoPro 20 May 2026/100GOPRO/Badminton BXH 3.MP4
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/GoPro 20 May 2026/100GOPRO/Badminton BXH 4.MP4
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Roora June 15/Video 10 - Badminton Multicourt/Badminton BXH 2.MP4

GX010141 ###

/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton - Boxhill - 17 June/GX010141.MP4

GX010137 ###

/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton - Boxhill - 17 June/GX010137.MP4

Boxhill ###

/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton - Boxhill - 17 June/GX010137.MP4
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton - Boxhill - 17 June/GX010138.MP4
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton - Boxhill - 17 June/GX010139.MP4
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton - Boxhill - 17 June/GX010140.MP4
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton - Boxhill - 17 June/GX010141.MP4
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton - Boxhill - 17 June/GX020140.MP4
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton Boxhill 19 June/GX010142.MP4
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton Boxhill 19 June/GX010143.MP4
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton Boxhill 19 June/GX010144.MP4

/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton Boxhill 19 June/GX010145.MP4
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton Boxhill 19 June/GX010146.MP4
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton Boxhill 19 June/GX010147.MP4
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton Boxhill 19 June/GX020144.MP4
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton Boxhill 19 June/GX020146.MP4
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton Boxhill 19 June/GX030146.MP4
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Roora June 15/Video 12 - Badminton Multicourt/Boxhill Doubles.MP4

GX020094 ###

/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 2 May 2026/100GOPRO/GX020094.MP4
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Roora June 15/Video 4 - Badminton/GX020094.MP4

LargeTest ###

/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Highlights Generator Data/FinalOutputOfVeryLargeTestVideo_GeminiGen2Prompt.mp4
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Highlights Generator Data/Tests/Compressed_VeryLargeTestVideo.mp4
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Highlights Generator Data/Tests/FinalOutputOfVeryLargeTestVideo_CoachPrompt.mp4
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Highlights Generator Data/Tests/OutputOfVeryLargeTestVideo.mp4
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Highlights Generator Data/Tests/OutputOfVeryLargeTestVideo2.mp4
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Training/Golden Labelled/LargeTestVideo.MP4

TestLarge ###

/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Highlights Generator Data/Tests/FinalOutputOfTestLargeVideo.mp4
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Highlights Generator Data/Tests/FinalOutputOfTestLargeVideo_GeminiGen2Prompt.mp4
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Highlights Generator Data/Tests/FinalOutputOfTestLargeVideo_GeminiGenPrompt.mp4
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Highlights Generator Data/Tests/FirstOutputOfTestLargeVideo.mp4
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Highlights Generator Data/Tests/OutputOfTestLargeVideo_CoachPrompt.mp4
/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Training/Golden Labelled/TestLargeVideo.MP4

42. assistant (2026-06-29T11:57:24.840Z)

I now have enough to build a definitive golden-name ? F: source-file mapping. The `Training/Golden Labelled/` folder is the authoritative set of named golden source videos. Let me update my todos and define the golden mapping precisely. Note: `mahadevpura-1.MP4` in the bucket (548MB) matches `Badminton 12 June 2026/mahadevpura-1.MP4` on F:.

Let me record the golden mapping and check whether the bucket's `mahadevpura_1.mp4` size (548827049) matches F:'s mahadevpura-1.MP4.

43. user (2026-06-29T11:57:32.494Z)

```
=== sizes of key mahadevpura files on F ===
11522892112     /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 12 June 2026/mahadevpura-0.MP4
2611863875      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 12 June 2026/mahadevpura-1.MP4
9227260503      /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 12 June
```

2026/mahadevpura-2.MP4
9227260503 /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Roora June 15/Video 3 - Badminton Multicourt/mahadevpura-2.MP4
196182380 /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Training/Golden Labelled/MahadevpuraSingles.MP4
2359252911 /mnt/f/[Khelsutra] GoPro Hero Black 12/KhelSutra_work/kushagra_june12_proxies/mahadevpura-0.MP4
548827049 /mnt/f/[Khelsutra] GoPro Hero Black 12/KhelSutra_work/kushagra_june12_proxies/mahadevpura-1.MP4
1943236941 /mnt/f/[Khelsutra] GoPro Hero Black 12/KhelSutra_work/kushagra_june12_proxies/mahadevpura-2.MP4

=== bucket mahadevpura_1.mp4 size = 548827049 (does any F mahadevpura match?) ===
/mnt/f/[Khelsutra] GoPro Hero Black 12/KhelSutra_work/kushagra_june12_proxies/mahadevpura-1.MP4
548827049 1781549534.000000 2026-06-15T18:52:14 .mp4 VIDEO

=== GX030094 sizes on F vs bucket GX030094_proxy=1429944539 ===
10611615335 /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 2 May 2026/100GOPRO/GX030094.MP4
10611615335 /mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Roora June 15/Video 6 - Badminton/GX030094.MP4

44. assistant (2026-06-29T11:57:36.812Z)

Very important finding: the bucket's `mahadevpura_1.mp4` (548827049 bytes) is actually a **proxy** (low-res), matching F:'s `kushagra_june12_proxies/mahadevpura-1.MP4` ? NOT a full-res source. Similarly `GX030094_proxy.mp4` (1.4GB) is a proxy, while F: has the full-res 10.6GB `GX030094.MP4`. So the bucket has proxies/derived clips, not the full golden originals.

This sharpens the migration gap: the golden full-res source videos are essentially all missing from GCS. Let me now build the definitive golden mapping JSON and check the background hash progress.

45. assistant (2026-06-29T11:58:06.773Z)

{
  "_note": "Authoritative golden source files live in 'Training/Golden Labelled/' (renamed copies) and/or the original GX0xxxxx capture folders. Maps the 15 golden names from manifests/golden.json to their F: full-res source path(s). GCS 'videos/' contains only proxies/derived clips, NOT these full-res sources (except none full-res).",
  "golden": [
    {"name": "adarsh_avi_singles", "f_primary": "/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Training/Golden Labelled/Adarsh and Avi.MP4", "size": 11445052772, "also": ["/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 1 June 2026/100GOPRO/GX010108.MP4"]},
    {"name": "kushagra_singles", "f_primary": "/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Training/Golden Labelled/KushagraYashAviFirstVideo.MP4", "size": 8058214679, "also": ["/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 22 May 2026/gBaddy/GX010106.MP4"]},
    {"name": "largetest_doubles", "f_primary": "/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Training/Golden Labelled/LargeTestVideo.MP4", "size": 11531882761, "also": ["/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 2 May 2026/100GOPRO/GX010093.MP4"]},
    {"name": "gbaaddy", "f_primary": "/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Training/Golden Labelled/gBaaddy20May2026.MP4", "size": 7075098663, "also": ["/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 22 May 2026/gBaddy/GX020107.MP4"]},
    {"name": "testlarge_short", "f_primary": "/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Training/Golden Labelled/TestLargeVideo.MP4", "size": 1510427943, "also": ["/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 4 May 2026/100GOPRO/GX030097.MP4"]},
    {"name": "mahadevpura_singles", "f_primary": "/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Training/Golden Labelled/MahadevpuraSingles.MP4", "size": 196182380, "also": ["/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/2026 March 6/100GOPRO/GX010053.MP4", "/mnt/f/Data/Initial GoPRo with Sports Recordings/GX010053.MP4"]},

    {"name": "mahadevpura_2", "f_primary": "/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 12 June 2026/mahadevpura-2.MP4", "size": 9227260503, "also": ["/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Roora June 15/Video 3 - Badminton Multicourt/mahadevpura-2.MP4"]},
    {"name": "GX010128", "f_primary": "/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 12 June 2026/GX010128.MP4", "size": null, "also": ["/mnt/f/[Khelsutra] GoPro Hero Black 12/KhelSutra_work/kushagra_june12_proxies/GX010128.MP4 (PROXY-sized .mp4)"]},
    {"name": "mahadevpura_1", "f_primary": "/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 12 June 2026/mahadevpura-1.MP4", "size": 2611863875, "note": "full-res 2.6GB; GCS videos/mahadevpura_1.mp4 (548MB) is the PROXY, equals F: KhelSutra_work/kushagra_june12_proxies/mahadevpura-1.MP4"},
    {"name": "GX030094", "f_primary": "/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 2 May 2026/100GOPRO/GX030094.MP4", "size": 10611615335, "also": ["/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Roora June 15/Video 6 - Badminton/GX030094.MP4"], "note": "GCS videos/GX030094_proxy.mp4 (1.4GB) is the PROXY, not this 10.6GB full-res"},
    {"name": "Badminton_BXH_2", "f_primary": "/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/GoPro 20 May 2026/100GOPRO/Badminton BXH 2.MP4", "size": 10925859511, "also": ["/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Roora June 15/Video 10 - Badminton Multicourt/Badminton BXH 2.MP4"]},
    {"name": "GX010141", "f_primary": "/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton - Boxhill - 17 June/GX010141.MP4", "size": null},
    {"name": "GX010137", "f_primary": "/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton - Boxhill - 17 June/GX010137.MP4", "size": null},
    {"name": "Boxhill_Doubles", "f_primary": "/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 4 May 2026/100GOPRO/GX010097.MP4", "size": 11532555534, "also": ["/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Roora June 15/Video 12 - Badminton Multicourt/Boxhill Doubles.MP4"]},
    {"name": "GX020094", "f_primary": "/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Badminton 2 May 2026/100GOPRO/GX020094.MP4", "size": 11533390543, "also": ["/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Roora June 15/Video 4 - Badminton/GX020094.MP4"]}
  ],
  "gcs_videos_objects": [
    {"obj": "gs://khelsutra-rally-corpus/videos/GX010142_1080p.mp4", "size": 767766440, "kind": "1080p-derived"},
    {"obj": "gs://khelsutra-rally-corpus/videos/GX010143_1080p.mp4", "size": 632948235, "kind": "1080p-derived"},
    {"obj": "gs://khelsutra-rally-corpus/videos/GX010147_1080p.mp4", "size": 659514891, "kind": "1080p-derived"},
    {"obj": "gs://khelsutra-rally-corpus/videos/GX030094_proxy.mp4", "size": 1429944539, "kind": "proxy"},
    {"obj": "gs://khelsutra-rally-corpus/videos/mahadevpura-1.MP4", "size": 548827049, "kind": "proxy"},
    {"obj": "gs://khelsutra-rally-corpus/videos/mahadevpura_1.mp4", "size": 548827049, "kind": "proxy(dup-name)"},
    {"obj": "gs://khelsutra-rally-corpus/videos/mahadevpura_smoke_180s.mp4", "size": 90194321, "kind": "smoke-clip"},
    {"obj": "gs://khelsutra-rally-corpus/videos/SHANAN/VID_1.mp4", "size": 691158901, "kind": "shanan"},
    {"obj": "gs://khelsutra-rally-corpus/videos/SHANAN/VID_2.mp4", "size": 2794800788, "kind": "shanan"},
    {"obj": "gs://khelsutra-rally-corpus/videos/SHANAN/VID_3.mp4", "size": 2627544458, "kind": "shanan"}
  ]
}

46. user (2026-06-29T11:58:07.845Z)

File created successfully at: /home/avidullu/projects/khelsutra-guru/f_drive_index/golden_map.json (file state is current in your context ? no need to Read it back)

47. user (2026-06-29T11:58:14.966Z)

```
Todos have been modified successfully. Ensure that you continue to use the todo list to track
your progress. Please proceed with the current tasks if applicable
```

48. assistant (2026-06-29T11:58:17.510Z)

```
Let me check the background hash progress and wait for it to advance. The confirm set is ~94
files. Let me monitor.
```

49. user (2026-06-29T11:58:20.170Z)

```
  [30/94] 0.20GB in 1.1s (178MB/s) GX010053.MP4
  [31/94] 2.32GB in 12.9s (179MB/s) Mom Jan 26 Singing.MP4
  [32/94] 6.76GB in 33.1s (204MB/s) GX010137.MP4
  [33/94] 5.77GB in 27.0s (214MB/s) GX010141.MP4
--- hashed count ---
hashed: 33 | errors: 0
```

50. assistant (2026-06-29T11:58:24.736Z)

```
Progressing well. 33/94 done at ~200 MB/s. The remaining files include many 11GB videos, so this
will take a while (the confirm set totals roughly 350-400GB). Rather than block, let me prepare
the final report-building script now so everything is ready when hashing completes. Let me look
at the backend hashing reference to confirm the MD5 algorithm matches.
```

51. user (2026-06-29T11:58:28.038Z)

```
=== look for backend/utils/hashing.py reference (read-only, outside drive) ===
./badminton-highlight-indexer/backend/utils/hashing.py
no backend dir at repo root
```

52. user (2026-06-29T11:58:29.720Z)

```
"""Canonical file-identity hashing for the indexer.

`video_id` is the MD5 of the *whole* video file. Historically each entrypoint
(`backend/main.py`, `run_eval.py`, `scripts/run_phase3_eval.py`,
`scripts/run_process_and_stitch.py`)
defined its own `get_file_md5` with a different read-chunk size (64KB / 8KB / 1MB).
The hash values agreed (same bytes hashed), but the duplicated code was a footgun:
any future change to one copy would silently diverge the DB key.

This module is the single source of truth. Import `compute_video_id` everywhere a
video needs to be keyed.
"""

from __future__ import annotations

import hashlib

# 64 KB matches the value used by the live pipeline (backend/main.py) before this
# was centralized. The chunk size does not affect the resulting digest ? it only
# bounds memory use while streaming large files.
_CHUNK_SIZE = 65536


def compute_video_id(filepath: str) -> str:
    """Return the MD5 hex digest of the whole file ? the canonical video_id.

    Streams the file in fixed-size chunks so multi-gigabyte recordings never load
    into memory at once.
    """
    hasher = hashlib.md5()
    with open(filepath, "rb") as f:
        for chunk in iter(lambda: f.read(_CHUNK_SIZE), b""):
```

```
                hasher.update(chunk)
        return hasher.hexdigest()

```

53. assistant (2026-06-29T11:58:33.413Z)

```
The canonical algorithm is confirmed: streaming MD5 of the whole file. My `md5sum`/`hashlib.md5`
full-file digest is byte-identical (chunk size doesn't affect the digest). My computed full MD5
== `compute_video_id`.

Now let me build the report-generation script. It will read all the checkpoint JSONs and produce
the three structured outputs. I'll make it robust to partial full_md5 (provisional ids for
unhashed uniques).
```

54. assistant (2026-06-29T11:59:38.772Z)

```
#!/usr/bin/env python3
"""Build master_video_index.json, dedup_report.md, gcs_migration_gap.md from
the checkpointed intermediate artifacts. Re-runnable; reflects however much
full-MD5 has completed so far (provisional ids for not-yet-hashed uniques).
READ-ONLY w.r.t. /mnt/f.
"""
import json, os, datetime
from collections import defaultdict

BASE = '/home/avidullu/projects/khelsutra-guru/f_drive_index'
videos = json.load(open(f'{BASE}/videos.json'))
size_groups = json.load(open(f'{BASE}/size_groups.json'))         # {size: [idx]}  >=2
partial = json.load(open(f'{BASE}/partial_sigs.json'))            # {idx: psig}
confirm_groups = json.load(open(f'{BASE}/confirm_groups.json'))   # {"size|psig":[idx]} >=2
full_md5 = json.load(open(f'{BASE}/full_md5.json')) if os.path.exists(f'{BASE}/full_md5.json')
else {}
ffmeta = json.load(open(f'{BASE}/ffprobe_meta.json')) if
os.path.exists(f'{BASE}/ffprobe_meta.json') else {}
golden_map = json.load(open(f'{BASE}/golden_map.json'))
```

---- GCS objects (parsed from full ls) ----

```
gcs_videos = golden_map['gcs_videos_objects']  # list of {obj,size,kind}
gcs_sizes = defaultdict(list)
gcs_basenames = {}
for o in gcs_videos:
    gcs_sizes[o['size']].append(o['obj'])
    gcs_basenames[os.path.basename(o['obj']).lower()] = o['obj']
```

---- golden name -> set of F primary/also paths ----

```
golden_paths = {}     # path -> golden_name
for g in golden_map['golden']:
    for p in [g.get('f_primary')] + g.get('also', []):
        if p:
            cp = p.split(' (')[0]  # strip annotation
            golden_paths[cp] = g['name']
```

---- Build video_id per file (full md5 if available, else provisional) ----

```
def file_video_id(i):
    fm = full_md5.get(str(i))
    if fm and not str(fm).startswith('ERROR'):
        return fm, False
    # provisional: size:<bytes>:p:<partialsig>
    v = videos[i]
    ps = partial.get(str(i))
    if ps:
        return f"size:{v['size_bytes']}:p:{ps}", True
```

```
    return f"size:{v['size_bytes']}", True
```

---- Cluster duplicates by full md5 where available ----

Within each confirm_group, if all members have full md5, cluster by md5.

If md5 missing, fall back to (size,psig) as the dup cluster key (provisional dup).

```
dup_clusters = defaultdict(list)   # cluster_key -> [idx]
member_of = {}                     # idx -> cluster_key
for key, idxs in confirm_groups.items():
    # try full md5 clustering
    md5s = {}
    for i in idxs:
        fm = full_md5.get(str(i))
        if fm and not str(fm).startswith('ERROR'):
            md5s.setdefault(fm, []).append(i)
        else:
            md5s.setdefault('PROV:' + key, []).append(i)
    for ck, members in md5s.items():
        if len(members) >= 2:
            for m in members:
                dup_clusters[ck].append(m)
                member_of[m] = ck
```

assign dup_group_id sequential

```
group_ids = {}
for n, ck in enumerate(sorted(dup_clusters.keys()), 1):
    group_ids[ck] = 'dg%03d' % n
```

canonical pick: earliest mtime, tie-break shortest path

```
def pick_canonical(members):
    return sorted(members, key=lambda i: (videos[i]['mtime'], len(videos[i]['path']),
videos[i]['path']))[0]

canonical_idx = {}
for ck, members in dup_clusters.items():
    canonical_idx[ck] = pick_canonical(members)
```

---- Build master index: one entry per UNIQUE video ----

Unique set = canonicals of dup clusters + all files not in any dup cluster.

```
in_dup = set(member_of.keys())
entries = []

def gcs_match(v):
    bn = os.path.basename(v['path']).lower()
    size = v['size_bytes']
    same_name = gcs_basenames.get(bn)
    same_size = gcs_sizes.get(size)
    in_gcs = bool(same_name) or bool(same_size)
    obj = same_name or (same_size[0] if same_size else None)
    return in_gcs, obj, bool(same_name), bool(same_size)

def make_entry(i, dup_group_id=None, is_canonical=True, redundant_paths=None):
    v = videos[i]
    vid, prov = file_video_id(i)
    m = ffmeta.get(str(i), {})
    ingcs, obj, by_name, by_size = gcs_match(v)
    return {
        'video_id': vid,
        'video_id_provisional': prov,
        'path': v['path'],
        'size_bytes': v['size_bytes'],
```

```
        'mtime': v['mtime_iso'],
        'ext': v['ext'],
        'duration_s': m.get('duration_s'),
        'fps': m.get('fps'),
        'width': m.get('width'),
        'height': m.get('height'),
        'codec': m.get('codec'),
        'dup_group_id': dup_group_id,
        'is_canonical': is_canonical,
        'redundant_copy_paths': redundant_paths or [],
        'in_gcs': ('yes' if ingcs else 'no'),
        'gcs_match_by_name': by_name,
        'gcs_match_by_size': by_size,
        'gcs_object': obj,
        'golden_name': golden_paths.get(v['path']),
    }
```

canonical entries for dup clusters

```
for ck, members in dup_clusters.items():
    ci = canonical_idx[ck]
    redundant = [videos[i]['path'] for i in members if i != ci]
    entries.append(make_entry(ci, group_ids[ck], True, redundant))
```

singleton (non-dup) entries

```
for v in videos:
    if v['idx'] not in in_dup:
        entries.append(make_entry(v['idx']))

entries.sort(key=lambda e: e['path'])
json.dump(entries, open(f'{BASE}/master_video_index.json', 'w'), indent=2)
```

================= dedup_report.md =================

stats from catalog

```
cat_counts = defaultdict(int); cat_bytes = defaultdict(int)
with open(f'{BASE}/catalog_raw.tsv', encoding='utf-8', errors='surrogateescape') as f:
    f.readline()
    for line in f:
        p = line.rstrip('\n').split('\t')
        if len(p) != 6: continue
        cat_counts[p[5]] += 1; cat_bytes[p[5]] += int(p[1])
total_files = sum(cat_counts.values()); total_bytes = sum(cat_bytes.values())

total_video_files = len(videos)
unique_videos = len(entries)
dup_copy_count = sum(len(m)-1 for m in dup_clusters.values())  # redundant copies
reclaimable = 0
for ck, members in dup_clusters.items():
    ci = canonical_idx[ck]
    for i in members:
        if i != ci:
            reclaimable += videos[i]['size_bytes']

confirmed_clusters = [ck for ck in dup_clusters if not ck.startswith('PROV:')]
prov_clusters = [ck for ck in dup_clusters if ck.startswith('PROV:')]

def human(b):
    for u in ['B','KB','MB','GB','TB']:
        if b < 1024 or u == 'TB':
            return f'{b:.2f} {u}'
        b /= 1024

lines = []
```

```
lines.append('# F: Drive Dedup Report (READ-ONLY ? report/plan only, NOTHING executed)')
lines.append('')
lines.append(f'_Generated: {datetime.datetime.now().isoformat()}_  ')
lines.append(f'_Canonical id = full-file MD5 (backend/utils/hashing.py::compute_video_id)._')
lines.append('')
lines.append('## Totals scanned')
lines.append(f'- Total media files: **{total_files}**')
lines.append(f'- Total bytes: **{total_bytes:,}** ({human(total_bytes)})')
for c in ('VIDEO','PROXY','THUMB','OTHER'):
    lines.append(f'  - {c}: {cat_counts[c]} files, {human(cat_bytes[c])}')
lines.append('')
lines.append('## Video dedup summary')
lines.append(f'- VIDEO files: **{total_video_files}**')
lines.append(f'- Unique videos (canonical): **{unique_videos}**')
lines.append(f'- Duplicate (redundant) copies: **{dup_copy_count}**')
lines.append(f'- Duplicate groups: **{len(dup_clusters)}** ({len(confirmed_clusters)} confirmed
by full-MD5, {len(prov_clusters)} provisional by size+partial-sig)')
lines.append(f'- **Total reclaimable bytes (if redundant copies removed): {reclaimable:,}
({human(reclaimable)})**')
lines.append('')
lines.append('## Method (staged, efficient on 1.2 TB)')
lines.append('1. Group VIDEO by exact byte size ? distinct size = unique (no hash).')
lines.append('2. Same-size groups -> partial signature md5(first 16MB ++ last 16MB) via seek
reads.')
lines.append('3. Surviving (size,partial-sig) collisions -> FULL md5sum to CONFIRM exact dup =
video_id.')
lines.append('')
lines.append('## Duplicate groups (PROPOSED canonical-to-keep = earliest mtime, tie-break
shortest path)')
lines.append('')
for ck in sorted(dup_clusters.keys(), key=lambda k: group_ids[k]):
    members = dup_clusters[ck]
    ci = canonical_idx[ck]
    gid = group_ids[ck]
    confirmed = not ck.startswith('PROV:')
    vid = full_md5.get(str(ci)) if confirmed else f'PROVISIONAL
size:{videos[ci]["size_bytes"]}:p:{partial.get(str(ci))}'
    gn = golden_paths.get(videos[ci]['path'])
    reclaim_grp = sum(videos[i]['size_bytes'] for i in members if i != ci)
    lines.append(f'### {gid}  ({"CONFIRMED full-MD5" if confirmed else "PROVISIONAL
size+partial-sig"})')
    lines.append(f'- video_id (full MD5): `{vid}`')
    lines.append(f'- size each: {videos[ci]["size_bytes"]:,} bytes
({human(videos[ci]["size_bytes"])})')
    if gn: lines.append(f'- golden_name: **{gn}**')
    lines.append(f'- KEEP (canonical): `{videos[ci]["path"]}`  (mtime
{videos[ci]["mtime_iso"]})')
    lines.append(f'- redundant copies ({len(members)-1}), reclaimable {human(reclaim_grp)}:')
    for i in members:
        if i != ci:
            lines.append(f'    - DELETE-CANDIDATE: `{videos[i]["path"]}`  (mtime
{videos[i]["mtime_iso"]})')
    lines.append('')
lines.append('## PROPOSED dedup action list (NOT EXECUTED ? review before any action)')
lines.append('The following redundant copies could be removed to reclaim '
             f'{human(reclaimable)}. **No deletions were performed.**')
lines.append('')
for ck in sorted(dup_clusters.keys(), key=lambda k: group_ids[k]):
    ci = canonical_idx[ck]
    for i in dup_clusters[ck]:
        if i != ci:
            lines.append(f'- [ ] rm -- {videos[i]["path"]!r}   # keep {videos[ci]["path"]!r}')
lines.append('')
lines.append(f'## Proxies / thumbs (informational, NOT dedup targets here)')
lines.append(f'- PROXY (.lrv): {cat_counts["PROXY"]} files, {human(cat_bytes["PROXY"])}')
```

```
lines.append(f'- THUMB (.thm): {cat_counts["THUMB"]} files, {human(cat_bytes["THUMB"])}')
open(f'{BASE}/dedup_report.md','w').write('\n'.join(lines))
```

================== gcs_migration_gap.md ==================

```
gl = []
gl.append('# GCS Migration Gap ? golden/corpus videos on F: vs gs://khelsutra-rally-
corpus/videos/')
gl.append('')
gl.append(f'_Generated: {datetime.datetime.now().isoformat()}_')
gl.append('')
gl.append('## Finding')
gl.append('All 15 golden clips have **labels + trajectories** in the bucket, but the bucket '
          '`videos/` folder contains only a handful of derived/proxy/smoke objects ? the '
          '**full-res golden source videos are essentially NOT migrated** (closes DATA_IN_GCS
§5).')
gl.append('')
gl.append('## gs://khelsutra-rally-corpus/videos/ inventory')
for o in gcs_videos:
    gl.append(f'- `{o["obj"]}` ? {o["size"]:,} B ? kind: {o["kind"]}')
gl.append('')
gl.append('## 15 golden clips: present on F: and migration status')
gl.append('')
gl.append('| golden_name | on F: (full-res source) | in GCS videos/ as full-res? | needs upload
|')
gl.append('|---|---|---|---|')
gcs_full_res_names = set()  # we determined NONE of the videos/ objects are full-res golden
sources
for g in golden_map['golden']:
    name = g['name']
    fp = g.get('f_primary')
    on_f = 'YES' if fp else 'NO'
    # is there a full-res object in GCS? proxies/derived don't count
    in_gcs_full = 'NO (only proxy/derived if any)'
    needs = 'YES' if fp else 'unknown'
    gl.append(f'| {name} | {on_f} | {in_gcs_full} | {needs} |')
gl.append('')
present = sum(1 for g in golden_map['golden'] if g.get('f_primary'))
gl.append(f'**{present}/15 golden clips have a full-res source on F:. '
          f'0/15 full-res golden sources are in gs://.../videos/. '
          f'=> {present} still need GCS upload.**')
gl.append('')
gl.append('## Per-golden F: source paths (for upload staging)')
for g in golden_map['golden']:
    gl.append(f'- **{g["name"]}** -> `{g.get("f_primary")}`'
              + (f'  (size {g["size"]:,})' if g.get('size') else '')
              + (f'  NOTE: {g["note"]}' if g.get('note') else ''))
gl.append('')
gl.append('## CRITICAL upload caveat (bracket glob)')
gl.append('`gcloud storage cp` treats `[...]` in a path as a character-class GLOB, so the '
          'bracketed source root `/mnt/f/[Khelsutra] GoPro Hero Black 12/...` will NOT upload '
          'directly ? it expands to nothing / errors. Stage through a bracket-free path.')
gl.append('')
gl.append('### Suggested bracket-free staging recipe (DO NOT RUN here ? owner-gated)')
gl.append('```bash')
gl.append('# 1. Create a bracket-free staging dir OUTSIDE /mnt/f (read-only on F:):')
gl.append('STAGE=$HOME/gcs_upload_stage; mkdir -p "$STAGE"')
gl.append('# 2. Hard-link or copy (copy across filesystems) each golden full-res into STAGE')
gl.append('#    with a clean canonical name. Example (copy, since F: is a different FS):')
gl.append('cp "/mnt/f/[Khelsutra] GoPro Hero Black 12/GoPro Recorded Videos/Training/Golden
Labelled/Adarsh and Avi.MP4" \\')
gl.append('   "$STAGE/adarsh_avi_singles.mp4"')
gl.append('# 3. Upload from the bracket-free staging path:')
gl.append('gcloud storage cp "$STAGE/adarsh_avi_singles.mp4" gs://khelsutra-rally-
corpus/videos/adarsh_avi_singles.mp4')
```

```
gl.append('# 4. Verify size/crc32c, then clean STAGE.')
gl.append('# Alternative: gcloud storage cp supports no-glob via copying the file after
rename;')
gl.append('# there is no reliable --no-glob flag, so always stage bracket-free.')
gl.append('```')
open(f'{BASE}/gcs_migration_gap.md','w').write('\n'.join(gl))

print('Outputs written.')
print('master_video_index.json entries (unique videos):', len(entries))
print('VIDEO files total:', total_video_files)
print('dup clusters:', len(dup_clusters), '(confirmed', len(confirmed_clusters), '/ prov',
len(prov_clusters),')')
print('redundant copies:', dup_copy_count)
print('reclaimable bytes:', reclaimable, human(reclaimable))
print('full_md5 hashed:', len([k for k,v in full_md5.items() if not
str(v).startswith('ERROR')]))
```

55. user (2026-06-29T11:59:39.847Z)

File created successfully at: /home/avidullu/projects/khelsutra-guru/f_drive_index/build_outputs.py (file state is current in your context ? no need to Read it back)

56. assistant (2026-06-29T11:59:41.727Z)

Now let me check the hash progress and wait for the confirm pass to finish.

57. user (2026-06-29T11:59:45.309Z)

```
  [32/94] 6.76GB in 33.1s (204MB/s) GX010137.MP4
  [33/94] 5.77GB in 27.0s (214MB/s) GX010141.MP4
  [34/94] 11.45GB in 56.2s (204MB/s) GX010108.MP4
hashed: 34
```

58. assistant (2026-06-29T11:59:49.151Z)

34/94. Let me wait efficiently using a monitor loop that checks until the confirm pass finishes. The remaining ~60 files include many 11GB videos ? roughly 350GB to read at ~200MB/s ? 30 min. Let me use the Monitor tool to wait for the background job.

59. assistant (2026-06-29T11:59:51.562Z)

I'll use a Bash background command with an `until` loop that exits when the confirm pass writes its DONE line, so I get a single notification when it completes.

60. user (2026-06-29T11:59:59.140Z)

Command running in background with ID: bhon4nb6e. Output is being written to: /tmp/claude-1000/-home-avidullu-projects-khelsutra-guru/4f364eb7-a600-4a2c-9b37-e5a313b6db9d/tasks/bhon4nb6e.output. You will be notified when it completes. To check interim output, use Read on that file path.

61. assistant (2026-06-29T12:00:01.883Z)

While waiting for the confirm pass, let me generate a preliminary version of the outputs so the artifacts exist and are checkpointed (they'll be regenerated once hashing completes). This also validates the build script.

62. user (2026-06-29T12:00:04.308Z)

```
Outputs written.
master_video_index.json entries (unique videos): 180
VIDEO files total: 220
```

```
dup clusters: 37 (confirmed 14 / prov 23 )
redundant copies: 40
reclaimable bytes: 254056638359 236.61 GB
full_md5 hashed: 35
```

63. assistant (2026-06-29T12:00:08.799Z)

```
The build script works. Preliminary: 220 VIDEO files ? 180 unique, 40 redundant copies, ~236 GB
reclaimable. Note dup clusters = 37 here (vs 41 size-groups) because the 4 three-file groups
collapse and clustering is by md5/provisional. Let me verify the math: 41 (size,psig) groups but
the report shows 37 dup clusters ? let me double check this isn't dropping any. Actually the
difference: confirm_groups has 41 keys; some may have split during md5 clustering if a full md5
differed. Let me verify no dup groups were lost incorrectly.
```

64. user (2026-06-29T12:00:14.127Z)

```
confirm_groups (size,psig >=2): 41 covering 86 files
fully-hashed groups that AGREE (true dup): 13
fully-hashed groups that SPLIT (false dup): 0
groups still partially hashed: 28
```

65. assistant (2026-06-29T12:00:22.538Z)

```
Excellent ? zero false dups so far: every fully-hashed (size, partial-sig) group is a confirmed
exact duplicate. The 37 vs 41 in the preliminary build is just because the 4 three-file groups
produce 1 cluster each (not the discrepancy I worried about) ? actually 41 groups should map to
41 clusters. Let me recheck: the build counted "dup clusters: 37" because for partially-hashed
groups it uses the `PROV:key` cluster key, and confirmed ones use the md5. The count should
still be 41. Let me investigate the count of 37.
```

66. user (2026-06-29T12:00:28.896Z)

```
clusters: 37
MIXED group (some hashed, some not): 196182380|fc364b1dc8927bad694c5c48fa227117 ['H', 'H', 'U']
-> temporarily under-clustered until full hash completes
MIXED group (some hashed, some not): 1881912767|211187e0496bdcf382757dc8326401a4 ['H', 'U'] ->
temporarily under-clustered until full hash completes
MIXED group (some hashed, some not): 1542282226|6e7c2ffa3d991125863c4279d83a49c9 ['H', 'U'] ->
temporarily under-clustered until full hash completes
MIXED group (some hashed, some not): 1243054145|59ed13a5c72d0b8b3692d177e306413a ['H', 'U'] ->
temporarily under-clustered until full hash completes
MIXED group (some hashed, some not): 11445052772|b782f3cf539c900913cca978d49f0138 ['H', 'U'] ->
temporarily under-clustered until full hash completes
```

67. assistant (2026-06-29T12:00:32.349Z)

```
That explains it ? it's a transient artifact of partial hashing: in mixed groups, the hashed
member and unhashed member temporarily get different cluster keys, dropping below the ?2
threshold. Once the confirm pass finishes (all 86 fully hashed), all 41 groups will cluster
correctly. The final rebuild will fix this. This is exactly why I rebuild after hashing
completes.

Let me wait for the confirm pass to finish. Let me check current progress.
```

68. user (2026-06-29T12:00:34.888Z)

```
  [35/94] 11.33GB in 57.6s (197MB/s) GX010128.MP4
  [36/94] 2.61GB in 13.2s (198MB/s) mahadevpura-1.MP4
hashed: 36
```

69. assistant (2026-06-29T12:00:40.797Z)

```
36/94. The remaining 58 files are mostly large (~11GB each), so this confirm pass needs to read
roughly 350-400GB more ? about 30-40 minutes at ~200MB/s. The waiter background job will notify
```

me when it completes. Let me let it run and not poll wastefully. I'll wait for the notification.

The confirm pass is the critical correctness step. Meanwhile, everything else is done: catalog, ffprobe, GCS cross-ref, golden mapping, and the build script all validated. Once the confirm pass finishes I'll rebuild outputs (fixing the transient under-clustering), then optionally start the progressive "all" full-MD5 pass on remaining uniques if time allows.

Let me wait for the waiter job notification.

70. user (2026-06-29T12:00:42.178Z)

<system-reminder>Warning: the file exists but is shorter than the provided offset (1). The file has 1 lines.</system-reminder>

71. assistant (2026-06-29T12:00:44.558Z)

The waiter is still running (no output yet, as expected). I'll wait for the completion notification rather than poll. Let me pause here and await the background job.

72. user (2026-06-29T12:00:47.479Z)

```
  [35/94] 11.33GB in 57.6s (197MB/s) GX010128.MP4
  [36/94] 2.61GB in 13.2s (198MB/s) mahadevpura-1.MP4
hashed: 36
```

73. assistant (2026-06-29T12:00:51.821Z)

Still at 36 (next file is a large 11GB one in progress). I'll await the waiter's completion notification. The system will re-invoke me when the confirm pass finishes.

I have completed all non-blocking work. The confirm full-MD5 pass is running in the background and a waiter will notify me on completion. Let me stop polling and wait for that event.

74. user (2026-06-29T12:00:53.045Z)

<system-reminder>Warning: the file exists but is shorter than the provided offset (36). The file has 1 lines.</system-reminder>

75. assistant (2026-06-29T12:00:59.278Z)

The output file shows differently than my tee'd copy. The hashing is running. I'll wait for the waiter background job (bhon4nb6e) to notify me when "DONE mode=confirm" appears. Pausing now to await that notification.

76. user (2026-06-29T12:01:03.881Z)

```
  [36/94] 2.61GB in 13.2s (198MB/s) mahadevpura-1.MP4
  [37/94] 9.23GB in 47.6s (194MB/s) mahadevpura-2.MP4
hashed: 37
```

77. assistant (2026-06-29T12:01:07.089Z)

Progressing (37/94). The pace is steady at ~195 MB/s. I'll now wait for the completion notification from the waiter job rather than continue polling, since the remaining work depends on the hash finishing. Awaiting notification.

78. user (2026-06-29T12:01:07.882Z)

<system-reminder>Warning: the file exists but is shorter than the provided offset (1). The file has 1 lines.</system-reminder>

79. assistant (2026-06-29T12:01:11.407Z)

The waiter is still polling. I'll wait for its completion notification. Let me stop here and await the background job event.

80. user (2026-06-29T12:01:14.552Z)

```
  [37/94] 9.23GB in 47.6s (194MB/s) mahadevpura-2.MP4
hashed: 37
```

81. assistant (2026-06-29T12:01:17.282Z)

I'll await the waiter notification now instead of repeatedly polling. The background job `bhon4nb6e` will report when the confirm full-MD5 pass completes.

82. user (2026-06-29T12:01:17.735Z)

<system-reminder>Warning: the file exists but is shorter than the provided offset (1). The file has 1 lines.</system-reminder>

83. assistant (2026-06-29T12:01:20.560Z)

Waiting for the confirm full-MD5 pass to complete (currently 37/94, ~195 MB/s, ~350GB of large files remaining ? 30 min). The waiter background job will notify me on completion. I'll pause polling and await that event.

84. user (2026-06-29T12:01:23.607Z)

```
  [37/94] 9.23GB in 47.6s (194MB/s) mahadevpura-2.MP4
hashed: 37
```

85. assistant (2026-06-29T12:01:30.203Z)

I'll stop polling and wait for the background waiter (bhon4nb6e) to fire its completion notification when the confirm pass finishes. Awaiting that event now.

86. user (2026-06-29T12:01:30.647Z)

<system-reminder>Warning: the file exists but is shorter than the provided offset (1). The file has 1 lines.</system-reminder>

87. assistant (2026-06-29T12:01:33.634Z)

The waiter is still running. I'll await its completion notification rather than continue manual polling. The confirm full-MD5 pass needs ~30 more minutes to read the remaining large files. I'll pause and wait for the system to notify me when the background job `bhon4nb6e` completes.